

Bullish-Bearish Strategies of Trading: A Non-linear Equilibrium *

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Abstract

In this paper, we study a financial market where risk neutral traders are endowed with a signal which is perfectly revealing of the direction (but not the exact amount) of the liquidation value of a normally distributed risky asset. This type of information is known as bullish or bearish and has been shown to have a great impact on prices. When the signal is positive (negative) the traders buy (sell) the asset. Trading recommendations issued by financial institutions fall into this category of information. At the equilibrium, the optimal strategy consists of buying (selling) when bullish (bearish) a fixed quantity of the asset. The market makers observe the aggregated volume submitted by the informed traders and the noise traders. In this model, since the optimal trading strategy is not linear, the pricing schedule is also a non-linear function of the volumes. We show i) that the price function is a non-linear Sigmoid-shaped function. ii) that price pressure is a regular bell-shaped curve of the volumes. iii) A monopolistic bullish-bearish type trader makes nearly thirty six percent of the profits she would have made with a perfect signal in a linear model à la Kyle (1985). iv) In the presence of competition, the market reveals his private information quicker than in a noisy informed strategic oligopoly and where the level of noise is chosen such that the monopoly bullish bearish expected profit is equal to the monopoly noisy informed expected profit with linearity. Moreover, the liquidity is no longer a monotonic increasing function of the number of competitors. This is true only for sufficiently large order imbalance leading to cross-liquidity-points for different size of the insider trading.

Keywords: Non-linear equilibrium, recommendations of trading, asymmetric information, investment banks.

JEL Classification: D43, D82, G14, G24.

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1 Introduction

The role of prices as a vector of information has been extensively studied in financial economics Grossman (1976), Grossman and Stiglitz (1980) and Kyle (1985). Most of the research in finance has focused on linear equilibrium. This is the case, in particular, in Kyle's (1985) seminal paper, Kyle (1989), Foster and Vishwanathan (1990), Holden and Subrahmanyam (1992), Vives (1995) among many other authors.

In the standard Kyle (1985) framework a monopolistic perfectly informed trader submits a market order to market makers. After having observed the total volume stemming from the informed and the noise traders, competitive market makers set the price in a Bayesian way. In this framework it is shown that there exists a unique linear equilibrium while assuming the linearity of prices. In this equilibrium, strategies are linear and the price function is a linear function of the aggregated volume *whatever the amount of the order imbalance*.

Most of the standard literature assumes that informed traders collect a perfect signal v Kyle (1984, 1985), Admati and Pfleiderer (1988a) or a noisy signal $s = v + \varepsilon$, Admati and Pfleiderer (1988b), Foster and Viswanathan (1994), Vives (1995), Dridi and Germain (1999) and as a result of the linear pricing schedule that the quantity bought is linear in this signal s . However, it seems to us that the most popular type of information is that of a **buy or a sell signal**. In effect, this type of information is well known from practitioners and is referred to as bullish versus bearish recommendations of trading. Most of the recommendations given by financial analysts or banks' research centers are bullish versus bearish signals. In order to understand how and why these predictions are built and to what extent these recommendations have tangible impact on prices in the market, we remind several important practices and studies.

- As a general rule and since it is conceptually impossible to have an exact prediction of the future value of any asset or project, the analysts summarize all their available information into what it is called a score function. For instance, Grinold (1994) studies the score which is a "standardized measure that shows how strongly you feel about a particular stock at a particular time". The score which is given by financial analysts or portfolio managers can take different values as "very positive", or "strong bullish", "very negative", or "strong bearish", "positive", or "bullish", "negative", or "bearish", "no view" or "hold" or simply "buy" or "sell" recommendations. In general, this measure statistically reflects the observed historical return on a particular stock, the accounting situation on a given firm, the forecasted returns and volatility and also the ability of the portfolio manager to interpret and forecast those data. More detailed recommendations by analysts can also be found such as buy till a certain pre-specified threshold.
- On the other hand, there is now an increasing empirical literature showing the importance of the impact of buy and sell recommendation on the stock prices. For instance,

Womack (1996) highlights that brokerage firms make enormous investments in building recommendations. He emphasizes on the ability of analysts to predict and influence stock prices. More precisely he analyzes the market reaction to new sell recommendation and to withdrawals of buy and sell recommendations. He documents significant initial price and volume reactions. Namely prices increase on average 3 percent for buy recommendations and drop 4.7 percent for sell recommendations. To illustrate the impact of this type of information, on the 27th of July 2000 the Financial Times published an article where it is mentioned that Amazon's stock fell as much as 9 % after a Lehman Brothers analyst had changed her "recommendation from a buy to a neutral" and another one "lowered his rating from strong buy to buy". Moreover Bjerring, Lakonishok and Vermaelen (1983) show how investors who follow the advice of a brokerage house would have earned positive abnormal returns. Barber and Loeffler (1993) use the analysts' recommendations published in the monthly Dartboard column of the Wall Street Journal. They show that the stocks selected by the analysts earn in excess of 4 percent abnormal returns for the 2 days after the publication in the Wall Street Journal and that average volume on the day of the publication is more than double normal volume levels.

- Finally Kim, Lin and Slovin (1997) stress that "the information about an analyst stock recommendation typically is given first to important clients of the brokerage firm and subsequently released to the general public. During the period between the pre-release of information and the public announcement, these clients possess private information and, thus, can be viewed as informed traders." In addition, according to the same authors "the Dow Jones News Wire reports the time the news is released to the public and indicates that the recommendation was disseminated to important clients of the brokerage firm before the market opening. As a result, there are typically several hours during which informed clients are able to act on the analyst recommendation before the news is publicly released."

To summarize the previous studies, the bullish versus bearish type of recommendation are commonly used on the market. They also have a major impact on prices. This type of information is first delivered to important clients who become informed traders before being released publicly. Thus it seems natural to consider the issue raised by bullish-bearish informed type of agents trading in a market.¹ In this paper we explore from the theoretical point of view the consequences of this type of collection of information on the prices. This corresponds to a genuine new problem that cannot be treated within the standard microstructure literature. We propose a theoretical framework to understand the price formation in such type of market information. Consequently when the ex-ante asset value is a gaussian random variable i) the strategy of

¹In this paper, however, we only study the case of "bullish", "hold" or "bearish" recommendations. One could always argue that what is more often observed on a financial market is a multi-level structure such as "strong buy", "buy", "hold", "sell", "strong sell" ; but we do believe that the simpler version "buy", "hold", and "sell" intrinsically contains the main relevant features of the phenomenon. We also want to stress that the computational burden for developing the theory within the framework of multi-level signals (this is notably the question of best response functions defined implicitly through elliptical forms) is such that we have decided to privilege the simpler version - namely "bullish" versus "bearish" signals.

trading of a “bullish versus bearish” trader is not linear. Hence, such an issue cannot be studied in the standard Kyle’s (1985) model because the price function is no longer a linear function of the observed volume.²

In light of these preliminary considerations, we study the issues raised by a “buy” or “sell” type of information, more precisely :

- What type of price function can we expect in such a market - which type of non-linearities is entailed by this type of information?
- In what extent is the use of this type of information optimal? In particular, when information is costly what would be the trade-off faced by a trader between acquiring v and bearing a large cost or getting only the direction of trading?³
- How quick does a competitive “bullish versus bearish” oligopoly reveal its information? What type of market efficiency is implied?

In the standard Kyle (1985) model the linearity of the price function holds essentially because of i) the linearity of the best response trading strategy of the informed agent with respect to his private information and ii) the normal distribution of the risky asset.

In the linear framework, the reactivity of the trader to a change in her private information is constant across the values of v . As a consequence, the price pressure - which reflects the change in the price following a change in the volume - is constant: a change in price is equal to $\lambda\Delta w$ where the positive parameter λ^{-1} indicates the market depth, and Δw is a measure of a net buy or a net sell order flow.

In a setting differing from the standard Kyle (1985) model where a monopolistic informed trader submits limit orders - or equivalently observes the realization of the noise trading - Rochet and Vila (1994) show the existence and the uniqueness regardless of the distribution of the risky asset and that of the noise trading.⁴ In another paper, Cho and El Karoui (1997) relax the assumption of normality and derive a quasi-linear equilibrium in the Kyle framework. However, in their setting they consider a Stackelberg game between the informed trader and the market makers, where the informed trader is seen as the leader of the game. Bhattacharya and Spiegel (1991) and Genotte and Leland (1990) consider other types of non-linearities e.g stemming from the hedging needs of the traders.

We take here another route. In our framework, the ex-ante distribution of the risky financial asset as well as the noise trading are normally distributed. On the other hand, the process of collection of information is different.

The main features of our model are as follows :

²It is important to notice that setting a linear price is not efficient in this framework.

³This depends of course on the profits made when trading with a buy or a sell signal.

⁴See Back (1992) for a continuous time version of the Kyle (1985) model and Rochet and Vila (1994).

- A monopolistic risk neutral informed trader observes a buy or sell signal S about the liquidation value v of the risky asset which is assumed to be normally distributed $\tilde{v} \rightsquigarrow \mathcal{N}(0, \sigma_v^2)$.⁵
- Uninformed noise traders submit normally distributed order $\tilde{u} \rightsquigarrow \mathcal{N}(0, \sigma_u^2)$ independent of $\tilde{v}$.
- The informed trader places orders $X(S)$ to risk neutral market makers who supply liquidity at prices equal to the expectation of the final value of the asset conditionally on the total order flow stemming from the informed trader and the uninformed noise traders. At the equilibrium, the informed buys a positive quantity γ^* after having observed a bullish signal (resp. sells a negative quantity $-\gamma^*$ after having observed a bearish signal.)

Then, we generalize the model to the case of strategic competitive traders observing a bullish versus bearish signal.

Our results are as follows :

1. In the case of a single informed trader:

- the equilibrium price function is an anti-symmetric Sigmoid-shaped function.
- The price pressure $\frac{\partial p}{\partial w}(w)$ curve is a regular bell-shaped one.
- The trader's trading profits amount nearly 36% of the profits made by a monopolistic informed trader knowing exactly the value of the liquidation of v in a linear set-up as in Kyle (1985)⁶.

2. In the case of a strategic oligopoly:

- The same Sigmoid-shaped price function and bell-shaped price pressure are also exhibited as a function of the number of competitors.
- In the presence of competition, the revelation of the private information is quicker than in the linear case and the expected individual profit (and the expected aggregated profit) are decreasing at an exponential rate.
- At the limit, the pricing schedule is a step-shaped function. The market makers are in that case able to infer the fully semi strong efficient price. Namely, if $S = 1$ then $p = p^* = \frac{2\sigma_v}{\sqrt{2\pi}}$ (resp. if $S = -1$ then $p = -p^* = -\frac{2\sigma_v}{\sqrt{2\pi}}$). Intuitively, the value $\frac{2\sigma_v}{\sqrt{2\pi}}$ corresponds to the mean of the truncated normal distribution on the positive real line. In other words, it is the value separating the positive real axis into two equally weighted regions of one half probability.

⁵Without any loss of useless generality we assume that the mean is zero. However, the extension to the case where the mean is different from zero is straightforward and does not change the results.

⁶The Kyle (1985) model is mentioned here just for sake of comparison.

- Moreover, the liquidity is no longer a monotonic increasing function of the number of competitors. This is true only for sufficiently large order imbalance leading to cross-liquidity-points for different size of the trading. This does correspond to new results that we will relate to the informational content of the submitted volumes in different oligopolies.

The predictions of our theoretical model are not in contradiction with the empirical findings (namely the non linear aspects of both price function and price pressure) such as in those of Kempf and Korn (1998) who analyze through a battery of parametric and robust non-parametric econometric methods the shape of the equilibrium price as a function of the order imbalance. However and in order to compare the predictions of our model to the evidence on the market, we would need studies that focuses on the link between prices and the nature of the collected information by the already mentioned informed traders. So far the empirical studies do not enable to identify the effects of the release of information. According to our model, the non-linearities of the shape of the price function should be enhanced by the bullish bearish type of information. It would therefore be interesting to attempt to disentangle such effects through some adapted testing procedure especially when there are data such as those referred by Kim, Lin and Slovin (1997) that enables to identify the precise time where the information is first given to some clients and then goes public.

The paper is organized as follows. We first lay out in section 2 the general set up and the considered model. We first study the monopoly case as a benchmark and then generalize the analysis to the case of many informed traders. We extensively analyze in section 3 the properties of the equilibrium (volumes, profits, liquidity, price pressure, price). Finally section 4 states some concluding remarks.

2 The Model

2.1 The monopoly case

We consider a financial market with a risky asset normally distributed $\tilde{v} \rightsquigarrow \mathcal{N}(0, \sigma_v^2)$. We denote the final liquidation value v . There are three types of agents:

1. A monopolistic risk neutral informed trader who observes in advance the direction of v i.e. he is endowed with a signal $\tilde{S}$ fully revealing of the bullish versus bearish state of the nature:

$$\begin{aligned} S &= -1 & \text{if } v < 0, \\ S &= 0 & \text{if } v = 0, \\ S &= 1 & \text{if } v > 0. \end{aligned}$$

2. Liquidity traders who submit market orders $\tilde{u} \rightsquigarrow \mathcal{N}(0, \sigma_u^2)$ independent of $\tilde{v}$.
3. Risk neutral market makers, who observe the aggregate volume $\tilde{w}$ and set rationally the price in a Bayesian way.

A strategy for the informed agent is a Lebesgue measurable function: $X : \{-1, 0, 1\} \longrightarrow \mathbb{R}$, determining his market order as a function of the observed signal S . For a given strategy X , let $\tilde{x} = X(\tilde{S})$. This strategy determines the aggregate order flow $\tilde{w} = \tilde{x} + \tilde{u}$.

Market makers observe the realization of the order flow, but not any of its components, and engage in a competitive auction. The outcome of this competition is described by a Lebesgue measurable function: $P : \mathbb{R} \longrightarrow \mathbb{R}$.

Given (P, X) , we denote $\tilde{p} = P(\tilde{w})$ and let $\tilde{\pi} = (\tilde{v} - \tilde{p})\tilde{x}$ the resulting trading profit of the insider.

The equilibrium conditions are that the competition in which market makers are involved drives their expected profits to zero conditional on the aggregate submitted volume and that the informed trader chooses his trading strategy so as to maximize his expected profit.

Definition 2.1 : $(P, X) \in L_2 \times L_2^7$ is an equilibrium if:

$$E[\tilde{v} - \tilde{p}/\tilde{w}] = 0, \quad (2.1)$$

and for all $X \in L_2$, given the (rational) beliefs of the market makers, and the corresponding price function $(P(\cdot))$, the informed chooses X to maximize his expected profits:

$$X \in \underset{x \in \mathbb{R}}{\operatorname{Argmax}} E[(\tilde{v} - P(x + \tilde{u}))x/S]. \quad (2.2)$$

Again, we recall here that we are in a strategic insider framework. So that the informed trader does take into account his impact onto prices. We are going to derive the unique perfect Bayesian equilibrium of this game.

Proposition 2.1 : *There exists a unique equilibrium (P, X) defined for:*

- the trading strategy $X(\cdot)$ by:

$$\begin{aligned} \tilde{x} = X(S) &= -\gamma^* = -\sigma_u, \text{ if } S = -1, \\ \tilde{x} = X(S) &= 0, \text{ if } S = 0, \\ \tilde{x} = X(S) &= \gamma^* = \sigma_u, \text{ if } S = 1. \end{aligned} \quad (2.3)$$

⁷ L_2 corresponds to the set of square integrable Lebesgue measurable functions.

- The equilibrium price function is:

$$\tilde{p} = P(w) = \frac{\sigma_v}{\sqrt{2\pi}} [K_2^*(w) - K_1^*(w)],$$

the functions $K_1^*(\cdot)$ and $K_2^*(\cdot)$ are defined as:

$$K_1^*(w) = \frac{2\varphi\left(\frac{w}{\sigma_u} + 1\right)}{\varphi\left(\frac{w}{\sigma_u} - 1\right) + \varphi\left(\frac{w}{\sigma_u} + 1\right)}, \quad (2.4)$$

$$K_2^*(w) = 2 - K_1^*(w),$$

where $\varphi(\cdot)$ corresponds to the probability distribution function of the standard normal variable $\mathcal{N}(0, 1)$.

Proof : See appendix A.

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The reaction function is equal to the level of noise trading measured by σ_u . The interpretation is again that the more the level of noise trading is and the more the informed trader can camouflage his trade. However it should be pointed out that, as an unexpected result in this non-linear framework, we still have a linear reaction with respect to the level of noise. Moreover, this model predicts that the informed trader submits medium size orders⁹. This is not the case in the standard literature since the informed can submit an order which is a linear function of his information and therefore unbounded.

It is possible to reinterpret the pricing schedule in terms of the probability that, market makers affect to the two states markov chain of nature (Bullish versus Bearish) and the associated semi strong efficient prices ($p^* = \frac{2\sigma_v}{\sqrt{2\pi}}$ if bullish and $-p^* = -\frac{2\sigma_v}{\sqrt{2\pi}}$ if bearish). Indeed if we define:

$$P_u(w) = \frac{K_2^*(w)}{2} = \frac{\varphi\left(\frac{w}{\sigma_u} - 1\right)}{\varphi\left(\frac{w}{\sigma_u} - 1\right) + \varphi\left(\frac{w}{\sigma_u} + 1\right)} = \frac{1}{1 + \exp\left(-\frac{2w}{\sigma_u}\right)}, \quad (2.5)$$

$$P_d(w) = \frac{K_1^*(w)}{2} = \frac{\varphi\left(\frac{w}{\sigma_u} + 1\right)}{\varphi\left(\frac{w}{\sigma_u} - 1\right) + \varphi\left(\frac{w}{\sigma_u} + 1\right)} = 1 - P_u(w),$$

then $0 < P_u(w) < 1$ corresponds to the probability that the market makers rationally assign to the event of a bullish state of the nature having observed the aggregate volume w . In this case

⁸We thank an anonymous referee who advises that in order to derive the equilibrium, we need to introduce both γ the informed trader's order and γ_c the conjecture that markets makers form about this order. At equilibrium both quantities are equal.

⁹This is consistent with the findings of Barclay and Warner (1993).

the semi strong efficient price would be p^* . Symmetrically, $P_d(w)$ corresponds to the probability that the market makers rationally assign to the event of a bearish state of the nature having observed the aggregate volume w . In the latter case the semi strong efficient price would be $-p^*$. In light of this, the price set by the market makers just corresponds to the expectation of the semi strong efficient ones with respect to the two states markov chain probability measure:

$$\tilde{p} = P_u(w)p^* + P_d(w)(-p^*). \quad (2.6)$$

In all the forthcoming graphs, we have normalized the parameters σ_u and σ_v to 1. The variable N corresponds to the number of informed traders and is equal to 1 in this subsection. The legend on the horizontal axis “volumes” refers to the aggregate submitted order both by the informed traders and the noise traders.

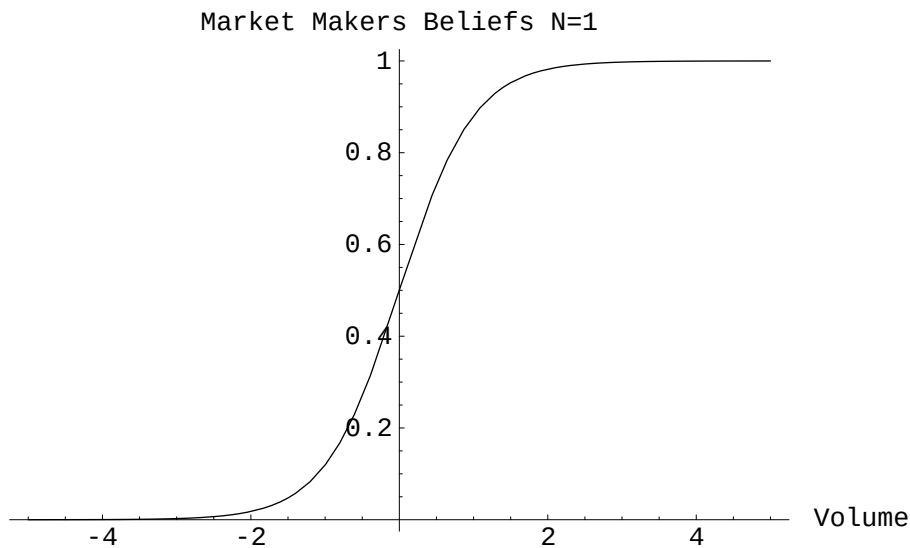


Figure 1: Market Makers Beliefs of the Bullish State $\mathbf{N} = 1$.

Thus we see that the probability that market makers affect for instance to the event of a bullish state of nature varies in a non-linear fashion. Indeed if the observed aggregate volume is 0 then we obtain a flat prior since $P_u(0) = \frac{1}{2}$. In other words market makers affect an equal probability of either a bullish or a bearish state of the final liquidation value. Then this probability increases from 0 to 1 when the aggregate volume vary from the negative to the positive real lines. The bigger the observed aggregate volume is and the higher is the affected probability to a bullish state. The same kind of symmetric comments can be stated for $P_d(w)$ which is related to the probability that market makers affect to a bearish state of nature after having observed the aggregate volume w .

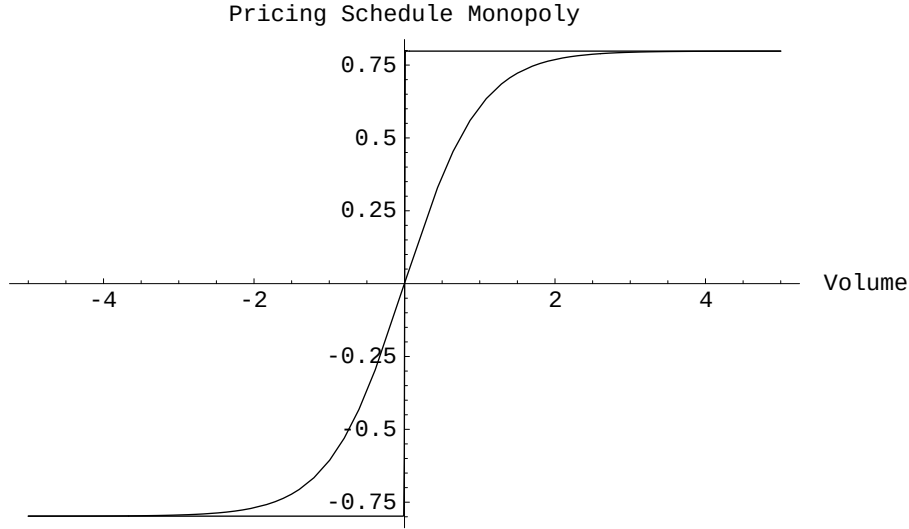


Figure 2: Pricing Schedule $\mathbf{N} = 1$.

As a consequence:

- the quoted price is always between the floor price $-p^*$ and the ceiling price p^* . The simple interpretation is that given that the informed agent knows in the worst case the bearish state of the risky asset, risk neutral market makers can just offer a maximum of p^* and quote $-p^*$. In the best case, the risk neutral informed agent knows a bullish state of the final liquidation value and only this and therefore is not prepared to pay more than p^* to acquire the risky asset,
- the price function is a Sigmoid-shaped curve (See figure 2). This is not in contradiction with the empirical evidence see Kempf and Korn (1998). The price pressure is not constant and decreases with the order imbalance. The two asymptotes $p = -p^*$ and $p = p^*$ are again explained by the probability the market makers affect to the event bullish state of nature after having observed the aggregate volume w .

Figures 3 and 4 represent respectively the liquidity and the price pressure $\frac{\partial \tilde{p}}{\partial w}$ as a function of the volumes in the case of one single informed trading agent in the market. We define naturally the **price pressure** as the derivative of the price function with respect to the volume. It indeed measures the price variation to a small change in the order imbalance. And the **liquidity** or market depth is defined as the inverse of the price pressure. This is in accordance with what has been previously introduced in the microstructure literature.

As already announced, both functions are not constant (As in Kyle (1985) linear equilibrium framework) and vary according to the volume. We see that the liquidity and price pressure are

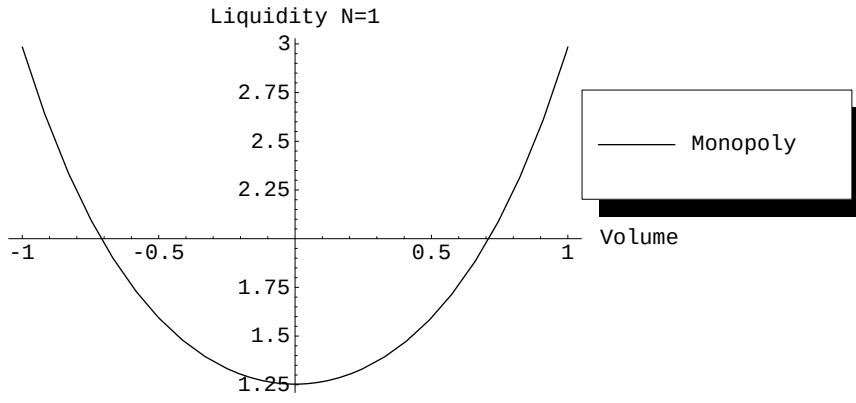


Figure 3: Liquidity $\mathbf{N} = 1$.

respectively U-shaped and bell-shaped. The price pressure (resp. the liquidity) is high (resp. small) for small volumes and it then decreases (resp. increases) with the order imbalance. In other words, the price function is marginally less sensitive for large aggregate volumes and a high (resp. low) level of liquidity is provided by market makers for large (resp. small) aggregated volumes. This suggests that the informational content of volumes (measured by the probability $P_u(w)$ affected by market makers to the event bullish state) is marginally less and less relevant when the absolute value of the aggregate volume is increasing.

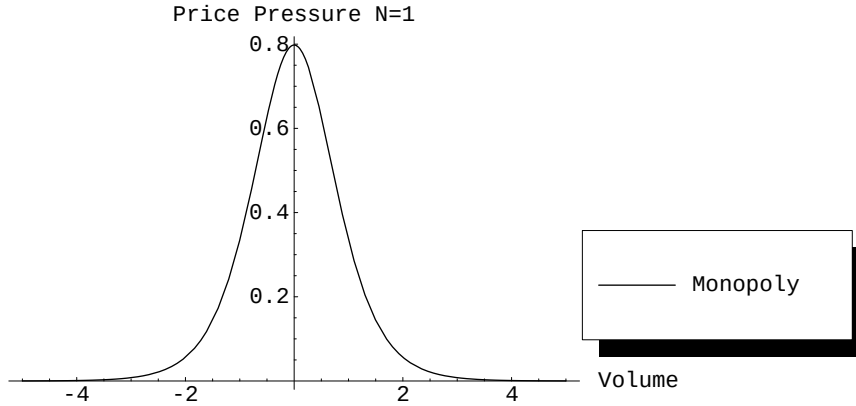


Figure 4: Price Pressure $\mathbf{N} = 1$.

The next Proposition studies the expected profit of a bullish-bearish trader with the one he achieves in the standard Kyle (1985).

Proposition 2.2 : *The expected profit $\pi^* = E(\tilde{\pi}^*)$ is:*

$$\begin{aligned}\pi^* &= \pi_{kyle} \times \xi^*, \\ \pi_{kyle} &= \frac{1}{2} \sigma_u \sigma_v, \\ \text{and } \xi^* &= \frac{4}{\pi} \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2)^2\right)}.\end{aligned}\tag{2.7}$$

Proof : *See appendix A.*

At the equilibrium, the numerical value, which corresponds to **a constant of the model**, is given by $\xi^* \simeq 0.36$. This implies that the reward for having collected the perfect information is equal to less than sixty four percent of the Kyle (1985) expected profits. One can wonder what are in practice the cost of a perfect information compared with those of having only the direction of trading.¹⁰ These expected profits can be achieved by a monopoly whose information precision $\frac{1}{\sigma_\varepsilon^2}$ would correspond to a value of σ_ε nearly equal to 2.59 times the original square root of the volatility σ_v .¹¹ Here again, as an unexpected result in this non-linear framework, we obtain that the profit is proportional as in Kyle (1985) to the product $\sigma_u \sigma_v$. The conditional expected profit while conditioning on the signal bullish versus bearish is equal to one half of the expected profit.

2.2 The oligopoly case

We generalize in this subsection the previous framework to the case where many informed traders are competing à la Nash in this market. More precisely, we consider a financial market with a risky asset normally distributed $\tilde{v} \rightsquigarrow \mathcal{N}(0, \sigma_v^2)$ and denote the final liquidation value v . Again the three types of agents are:

1. N risk neutral informed traders who observe in advance the direction of v i.e. they are each endowed with a signal $\tilde{S}$ fully revealing of the bullish-bearish state of the nature:

$$\begin{aligned}S &= -1 & \text{if } v < 0, \\ S &= 0 & \text{if } v = 0, \\ S &= 1 & \text{if } v > 0.\end{aligned}$$

2. Liquidity traders who submit market orders $\tilde{u} \rightsquigarrow \mathcal{N}(0, \sigma_u^2)$ independent of $\tilde{v}$.
3. Risk neutral market makers, who observe the aggregate volume $\tilde{w}$ and set rationally the price in a Bayesian way.

¹⁰Obviously, we can only compare the expected profits of an informed agent in the Kyle linear setup.

¹¹We define the precision of the signal $\tilde{S} = \tilde{v} + \tilde{\varepsilon}$ as the inverse of the variance of the noise $\tilde{\varepsilon}$: $\frac{1}{\sigma_\varepsilon^2}$ see Admati and Pfleiderer (1988b).

A strategy for the informed agent $i = 1, \dots, N$ is a Lebesgue measurable function: $X_i : \{-1, 0, 1\} \rightarrow \mathbb{R}$, determining his market order as a function of the observed signal S . For given strategies $(X_1, \dots, X_N)$, let $\tilde{x}_i = X_i(\tilde{S})$. These strategies determine the aggregate order flow $\tilde{w} = \sum_{i=1}^N \tilde{x}_i + \tilde{u}$.

Market makers observe the realization of the order flow, but not any of its components, and engage in a competitive auction. The outcome of this competition is described by a Lebesgue measurable function: $P : \mathbb{R} \rightarrow \mathbb{R}$.

Given $(P, X_1, \dots, X_N)$, we denote $\tilde{p} = P(\tilde{w})$ and let $\tilde{\pi}_i = (\tilde{v} - \tilde{p}) \tilde{x}_i$ the resulting trading profit of each insider $i = 1, \dots, N$.

The equilibrium conditions are that the competition in which market makers are involved drives their expected profits to zero conditional on the aggregate submitted volume and that the informed traders choose their trading strategies so as to maximize their expected profits.

Definition 2.2 : $(P, X_1, \dots, X_N) \in L_2^{N+1}$ is an equilibrium if:

$$E[\tilde{v} - \tilde{p}/\tilde{w}] = 0, \quad (2.8)$$

and for all $X_i \in L_2$, given the (rational) beliefs of the market makers, and the corresponding price function $(P(\cdot))$, each informed chooses X_i to maximize his expected profits:

$$X_i \in \underset{x \in \mathbb{R}}{\operatorname{Argmax}} E \left[\left(\tilde{v} - P(x + \sum_{j \neq i} \tilde{x}_j + \tilde{u}) \right) x / S \right]. \quad (2.9)$$

We derive the unique perfect Bayesian equilibrium of this game.

Proposition 2.3 : *There exists a unique equilibrium $(P, X_1, \dots, X_N)$ defined for:*

- the trading strategies $X_i(\cdot), i = 1, \dots, N$ by:

$$\begin{aligned} \tilde{x}_i &= X_i(S) = -\gamma^*(N) = -\frac{\sigma_u}{\sqrt{N}}, \text{ if } S = -1, \\ \tilde{x}_i &= X_i(S) = 0, \text{ if } S = 0, \\ \tilde{x}_i &= X_i(S) = \gamma^*(N) = \frac{\sigma_u}{\sqrt{N}}, \text{ if } S = 1. \end{aligned} \quad (2.10)$$

Therefore the equilibrium is a symmetric one, each informed agent chooses the same trading strategy and submits the same volume.

- The equilibrium price function is:

$$\tilde{p} = P(w) = \frac{\sigma_v}{\sqrt{2\pi}} [K_2^*(w, N) - K_1^*(w, N)],$$

the functions $K_1^*(\cdot, \cdot)$ and $K_2^*(\cdot, \cdot)$ are defined as:

$$K_1^*(w, N) = \frac{2\varphi\left(\frac{w}{\sigma_u} + \sqrt{N}\right)}{\varphi\left(\frac{w}{\sigma_u} - \sqrt{N}\right) + \varphi\left(\frac{w}{\sigma_u} + \sqrt{N}\right)}, \quad (2.11)$$

$$K_2^*(w, N) = 2 - K_1^*(w, N),$$

where $\varphi(\cdot)$ corresponds to the probability distribution function of the standard normal variable $\mathcal{N}(0, 1)$.

Proof : See appendix B.

The reaction function is again , as in the standard Kyle (1985), proportional to the level of noise trading measured by σ_u : the more the level of noise trading is and the more the informed trader can camouflage his trade.

It is also possible to reinterpret the pricing schedule in terms of the probability that, market makers affect to the two states markov chain of nature (Bullish versus Bearish) and the associated semi strong efficient prices ($p^* = \frac{2\sigma_v}{\sqrt{2\pi}}$ if bullish and $-p^* = -\frac{2\sigma_v}{\sqrt{2\pi}}$ if bearish). The probabilities are here:

$$P_u(w, N) = \frac{K_2^*(w, N)}{2} = \frac{\varphi\left(\frac{w}{\sigma_u} - \sqrt{N}\right)}{\varphi\left(\frac{w}{\sigma_u} - \sqrt{N}\right) + \varphi\left(\frac{w}{\sigma_u} + \sqrt{N}\right)} = \frac{1}{1 + \exp\left(-\frac{2w\sqrt{N}}{\sigma_u}\right)}, \quad (2.12)$$

$$P_d(w, N) = \frac{K_1^*(w, N)}{2} = \frac{\varphi\left(\frac{w}{\sigma_u} + \sqrt{N}\right)}{\varphi\left(\frac{w}{\sigma_u} - \sqrt{N}\right) + \varphi\left(\frac{w}{\sigma_u} + \sqrt{N}\right)} = 1 - P_u(w, N),$$

The price set by the market makers just corresponds to the expectation of the semi strong efficient ones with respect to the two states markov chain probability measure:

$$\tilde{p} = P_u(w, N)p^* + P_d(w, N)(-p^*). \quad (2.13)$$

Figure 5 represents the market makers beliefs about a bullish state of the risky asset for the monopoly (dotted line) and the duopoly (dash line) cases. The same patterns are confirmed

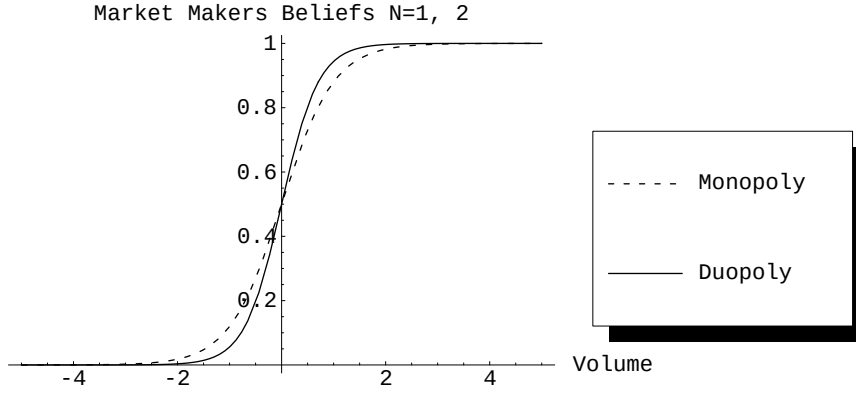


Figure 5: Market Makers Beliefs of the Bullish State $\mathbf{N} = 1, 2$.

in the oligopoly case. The probability that market makers affect for instance to the event of a bullish state of nature varies in a non-linear fashion. If the observed aggregate volume is 0 then we obtain a flat prior since $P_u(0, N) = \frac{1}{2}$ whatever the degree of competition in the market. This probability increases from 0 to 1 when the aggregate volume vary from the negative to the positive real lines. However the effect of competition is that the strength of the beliefs of the market makers is increasing with the number N of competitors. Thus we see that for a given positive volume the probability affected by the market makers to the bullish state event is increasing with N (see figures 5 – 6). This is essentially due to the fact that although the individual orders $\gamma^*(N)$ is decreasing with N the aggregate informed order $N\gamma^*(N)$ is increasing with N .

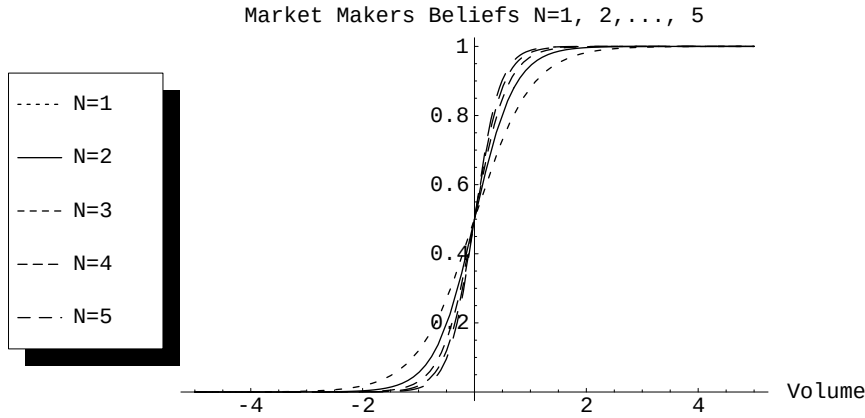


Figure 6: Market Makers Beliefs of the Bullish State $\mathbf{N} = 1, \dots, 5$.

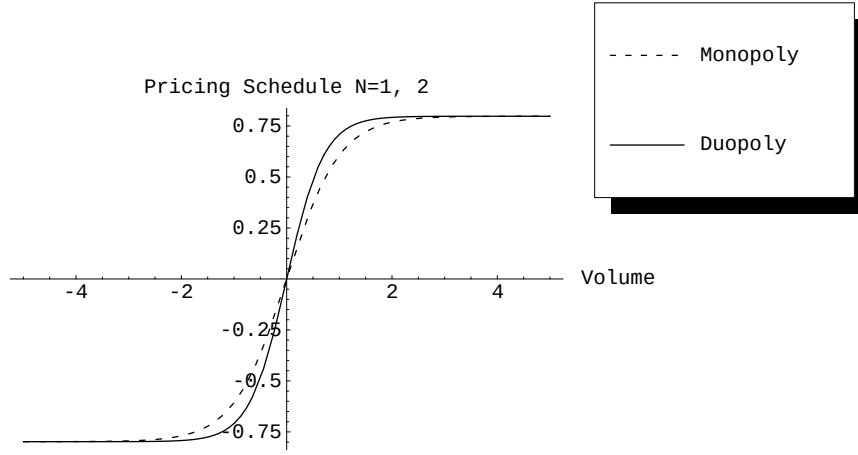


Figure 7: Pricing Schedule $\mathbf{N} = 1, 2$.

Figure 7 represents the pricing schedule both in the monopoly (dotted line) and the duopoly (dash line) cases. We still observe Sigmoid-shaped curves. However, we see that competition brings about more efficient prices since the dash line is above (resp. below) the dotted line for positive volumes (resp. negative volumes). The efficient pricing schedule would correspond to the step wise function ($p = -p^* = -\frac{2\sigma_v}{\sqrt{2\pi}}$ if $S = -1$ and $p = p^* = \frac{2\sigma_v}{\sqrt{2\pi}}$ if $S = 1$). This has again to be related to the treatment of the informational content of the aggregate volume by the market makers - through the probability they affect to the event of a bullish state of nature.

Proposition 2.4 : *The expected profit of each agent $\pi_i^*(N) = E(\tilde{\pi}_i^*)$, $i = 1, \dots, N$, is given by:*

$$\pi_i^*(N) = \frac{2\sigma_u\sigma_v}{\pi} \frac{1}{\sqrt{N}} \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t + 2\sqrt{N})^2\right)}. \quad (2.14)$$

Proof : *See appendix B.*

Figure 8 shows the expected individual profit expressed in fraction of the standard Kyle (1985) monopoly profit.¹² As a standard result, competition destroys the expected individual profit which goes to zero when the number of informed agents is large. This result is similar as the one we could expect to obtain in a Cournot model.¹³ The expected profits are again proportional to the product $\sigma_u\sigma_v$.

¹²For instance the individual profit in the monopoly case when bullish bearish signals are observed is equal to 0.36 times the standard $\frac{1}{2}\sigma_u\sigma_v$.

¹³However in Dridi and Germain (1999) we show that this is not always the case when the traders collect noisy information.

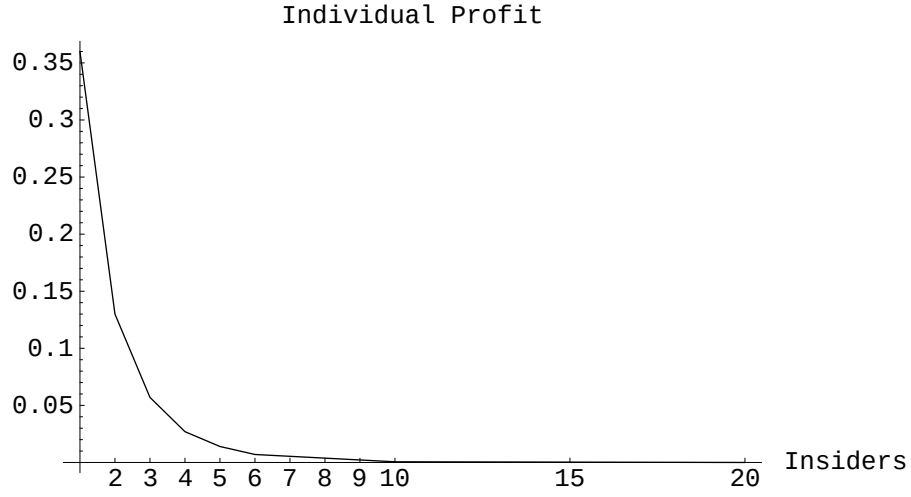


Figure 8: Individual Expected Profit.

3 Properties of the equilibrium

3.1 Volumes

- At the equilibrium, the individual order $\gamma^*(N)$ (resp. $-\gamma^*(N)$) is a decreasing (resp. increasing) function of the number N of informed agents when a bullish (resp. bearish) signal S is observed.
- The aggregated volumes $N\gamma^*(N)$ (resp. $-N\gamma^*(N)$) is an increasing (resp. decreasing) function of the number of informed traders N when a bullish (resp. bearish) signal S is observed.

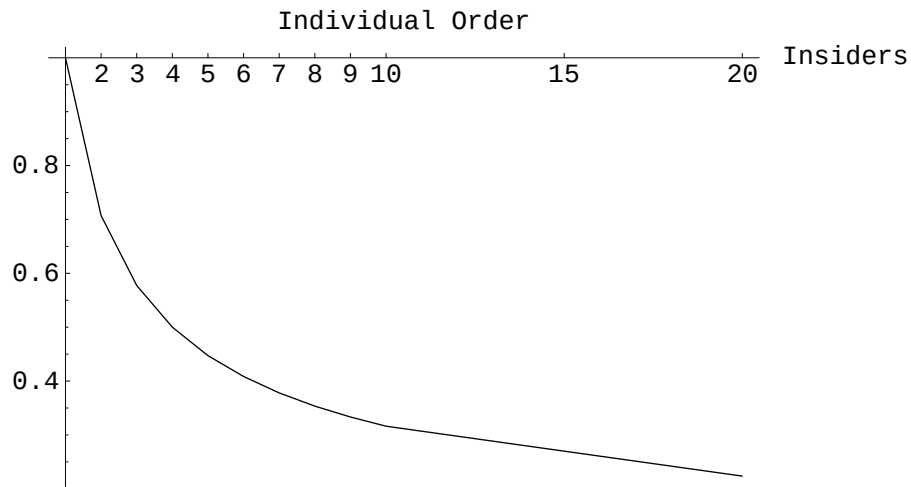


Figure 9: Individual Order N-Oligopoly: Bullish Signal.

Figure 9 represents the individual order in the N-oligopoly case and when a bullish signal $S = 1$ has been observed. Indeed, the bearish case is deduced from the previous case just by a symmetry with respect to the horizontal axis. The individual order is a decreasing function of the number of insiders. In effect, each informed trader endogenizes the reaction of the other insiders. This leads to a lessened individual reaction. The rate of convergence to zero is $\frac{1}{\sqrt{N}}$.



Figure 10: Aggregate Order N-Oligopoly: Bullish Signal.

Figure 10 represents the aggregate orders in the N-oligopoly case and when a bullish signal $S = 1$ has been observed. The aggregate volume is increasing with the number of insiders and tends to infinity at the rate $\sqrt{N}$.

3.2 Expected profits

Proposition 3.1 : *At the equilibrium, the expected individual profit $\pi_i^*(N)$ is a decreasing function of the number N of informed agents and we have the following equivalent when N is large:*

$$\pi_i^*(N) \underset{N \rightarrow +\infty}{\sim} \frac{\sigma_u \sigma_v}{N} \exp\left(-\frac{N}{2}\right). \quad (3.1)$$

Proof : *See appendix C.*

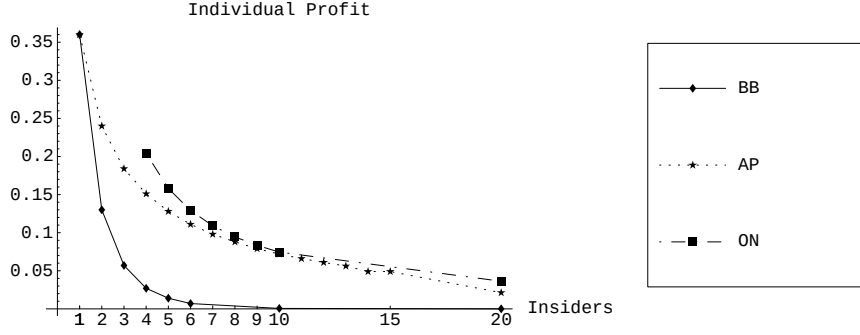


Figure 11: Individual Expected Profit: a Comparison.

In figure 11, we have plotted the expected individual profit as a function of the number of informed traders for three different models:

- bullish-bearish oligopoly: dash line (BB),
- noisy informed oligopoly (with linear equilibrium) as developed by Admati and Pfleiderer (1988b) and where the choice of the variance of the noise in the signal σ_ε^2 is such that the monopoly in this case achieves exactly thirty six percent of the Kyle (1985) monopoly profits- this value corresponds of course to the monopoly profits in case of bullish bearish signals: dotted line - star points (AP),
- optimally noisy oligopoly (with linear equilibrium) as developed by Dridi and Germain (1999)¹⁴: dotted line - square points ($N \geq 4$ ON).

Figure 11 shows that, the individual expected profit goes to zero even faster in the case of the bullish-bearish oligopoly when the number of competitors in the market is increasing. The rate of convergence is indeed an exponential one $\frac{1}{N} \exp\left(-\frac{N}{2}\right)$.

Proposition 3.2 : *At the equilibrium, the expected aggregated profits $N\pi_i^*(N)$ is a decreasing function of the number N of informed agents and we have the following equivalent when N is large:*

$$N\pi_i^*(N) \underset{N \rightarrow +\infty}{\sim} \sigma_u \sigma_v \exp\left(-\frac{N}{2}\right). \quad (3.2)$$

Proof : *See appendix C.*

¹⁴Dridi and Germain (1999) have shown that with linear equilibrium and noisy signals with heterogeneous beliefs, there exists an optimal level of noise in the signals which maximizes each individual profit for $N \geq 4$.

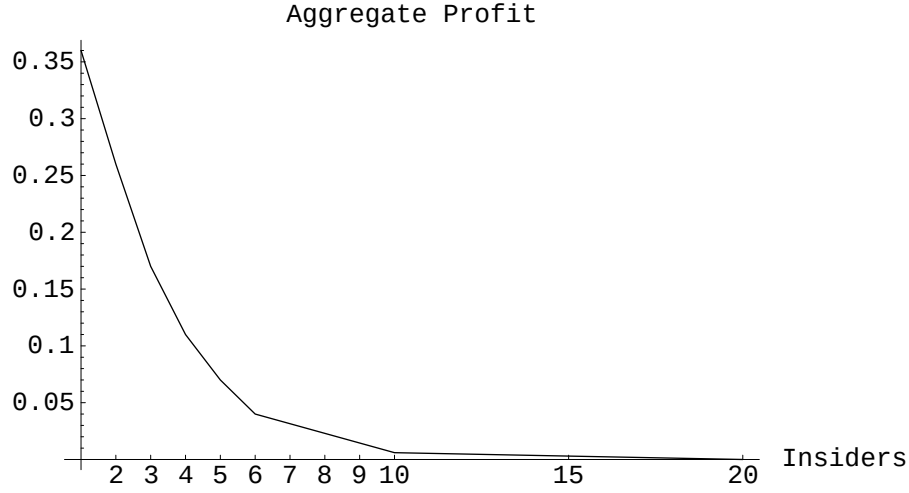


Figure 12: Aggregate Expected Profit N.

Figures 12 and 13 show that, the expected aggregate profit¹⁵ in the bullish-bearish setup (dash line) is a decreasing function of the number of insiders. Moreover in this case, it goes even faster to zero when the number of informed traders is large in comparison with the noisy informed (dotted lines, star points) framework as developed by Admati and Pfleiderer (1988b) and where the choice of the variance of the noise in the signal σ_ε^2 is such that the monopoly in this case achieves exactly thirty six percent of the Kyle (1985) monopoly profits and the optimally noisy framework (dotted line, square points). This might be interpreted, all other things being equal, by a quicker revelation of information. It is worth mentioning that in the AP framework, Dridi and Germain (1999) have shown that the aggregate profits are first increasing with the number of insiders and then decreasing for any given fixed level of noise in the signals. This leads to define an optimal size of the market in this case 14.

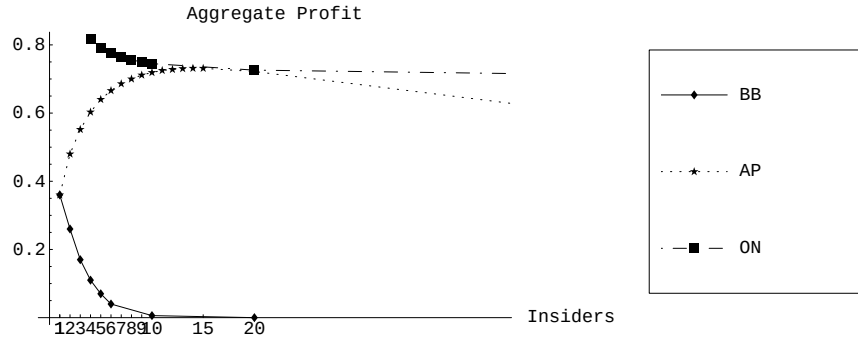


Figure 13: Aggregate Expected Profit: a Comparison.

¹⁵Again expressed as a fraction of the standard Kyle (1985) expected profits $\frac{1}{2}\sigma_u\sigma_v$.

3.3 The liquidity

Proposition 3.3 : *The liquidity $\lambda^{*-1}(w, N)$ is defined by $\lambda^*(w, N) = \frac{\partial \tilde{p}}{\partial w}$ and is equal to:*

$$\lambda^*(w, N) = \frac{8\sigma_v}{\sqrt{2\pi}\sigma_u} \frac{\sqrt{N} \exp\left(\frac{2w\sqrt{N}}{\sigma_u}\right)}{\left[1 + \exp\left(\frac{2w\sqrt{N}}{\sigma_u}\right)\right]^2}. \quad (3.3)$$

Moreover when N is large, the liquidity and the pricing schedule are, for any given level of noise trading u , respectively equivalent to:

$$\begin{aligned} \lambda^*(w, N) &\underset{N \rightarrow +\infty}{\sim} \frac{8\sigma_v}{\sqrt{2\pi}\sigma_u} \sqrt{N} \exp(-2N) \exp\left(-\frac{2u}{\sigma_u} \sqrt{N}\right), \quad \text{if } S = 1, \\ \lambda^*(w, N) &\underset{N \rightarrow +\infty}{\sim} \frac{8\sigma_v}{\sqrt{2\pi}\sigma_u} \sqrt{N} \exp(-2N) \exp\left(\frac{2u}{\sigma_u} \sqrt{N}\right), \quad \text{if } S = -1, \\ \tilde{p} &\underset{N \rightarrow +\infty}{\sim} \frac{2\sigma_v}{\sqrt{2\pi}} \quad \text{if } S = 1, \\ \tilde{p} &\underset{N \rightarrow +\infty}{\sim} -\frac{2\sigma_v}{\sqrt{2\pi}} \quad \text{if } S = -1. \end{aligned} \quad (3.4)$$

Proof : See appendix C.

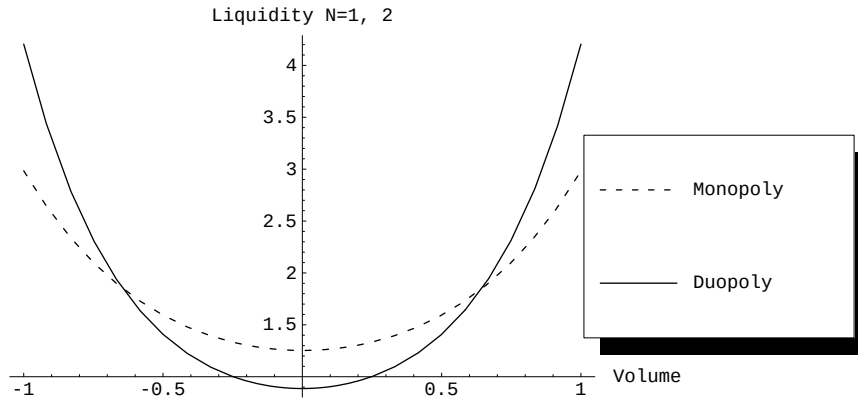


Figure 14: Liquidity $\mathbf{N} = 1, 2$ Small Volumes.

Figures 14 and 15 represent the liquidity as a function of the volumes for the monopoly (dotted line) and the duopoly (dash line) and respectively for small and large volumes. We observe that,

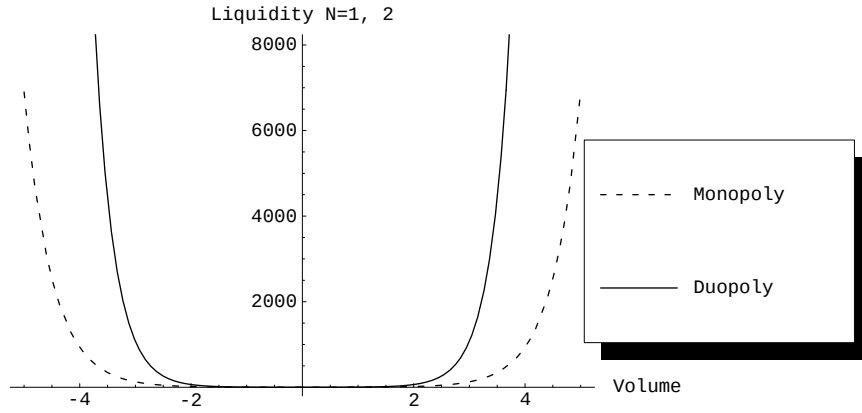


Figure 15: Liquidity $\mathbf{N} = 1, 2$ Large Volumes.

while for small volumes, the liquidity is higher for the monopoly case ($N=1$), for larger volumes the liquidity is higher in the duopoly case.

These patterns are confirmed for the general case (see figure 16), namely that for small volumes the liquidity is decreasing between two consecutive liquidity curves and for larger volumes it increases. However, it should be stressed that, the region (small volumes), for which the liquidity is decreasing between two consecutive liquidity curves, is lessening with the number of insiders.

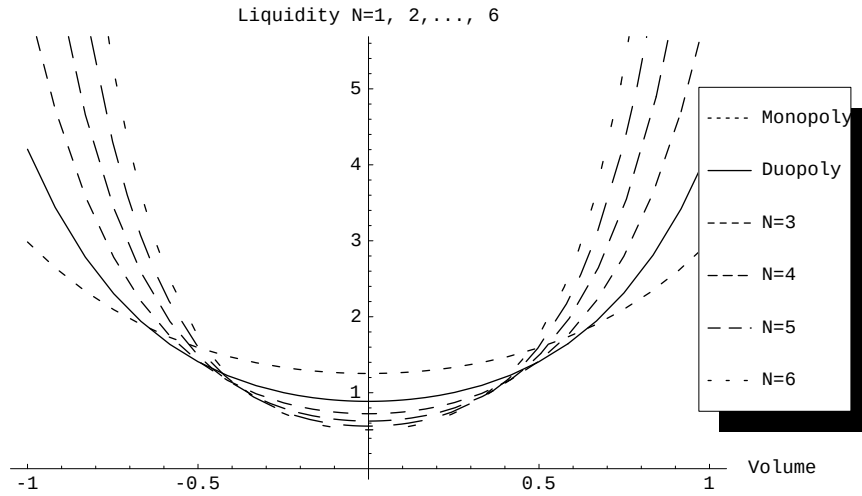


Figure 16: Liquidity $\mathbf{N}$.

Figures 17 and 18 represent the price pressure as a function of the volumes. Since the price pressure just corresponds to the inverse of the liquidity, analogous results as for the liquidity

can be stated. One possible way to understand those patterns is to examine the marginal value of the information brought about by the aggregate volume observed by the market makers. Indeed the price pressure, which measures the intensity of the adjustment of prices to the volume, is first higher for the oligopoly in comparison with the monopoly (or more generally with any oligopoly of less informed competitors) because for those ranges the marginal value of the information and therefore the adjustment is higher in the biggest oligopoly. This is also explained by the respective marginal variation in the probabilities that market makers affect to the bullish state of the nature. In other words, the oligopoly of a higher number of informed traders adjust its prices more sharply when there is a small amount of information in the trades.

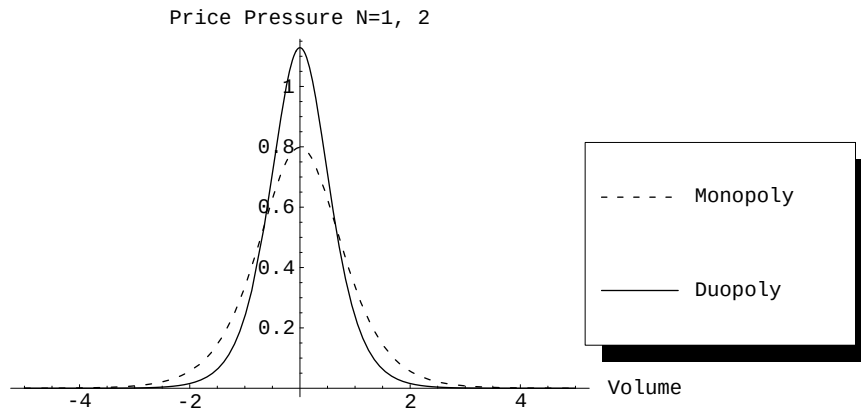


Figure 17: Price Pressure $N = 1, 2$.

Figure 19 represents the different pricing schedule corresponding to the different oligopolies N as a function of the volume.

In figure 20, we have plotted the pricing schedule in the bullish state of nature as a function of the noise trading given that, at equilibrium, the order $\gamma^*(1)$ submitted by the informed agent is known. In terms of welfare implications, we see that as long as the noise trading is not too strongly in the opposite direction of the state of nature (here bullish), then the price is close

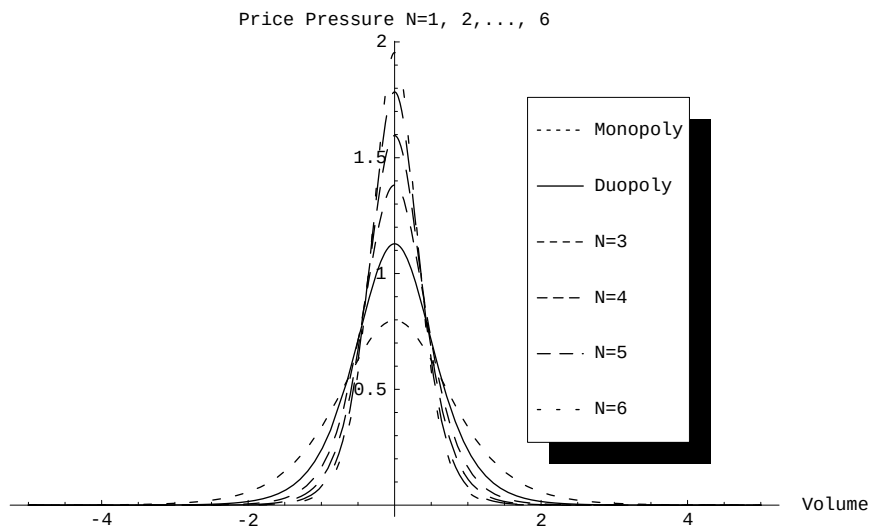


Figure 18: Price Pressure N .

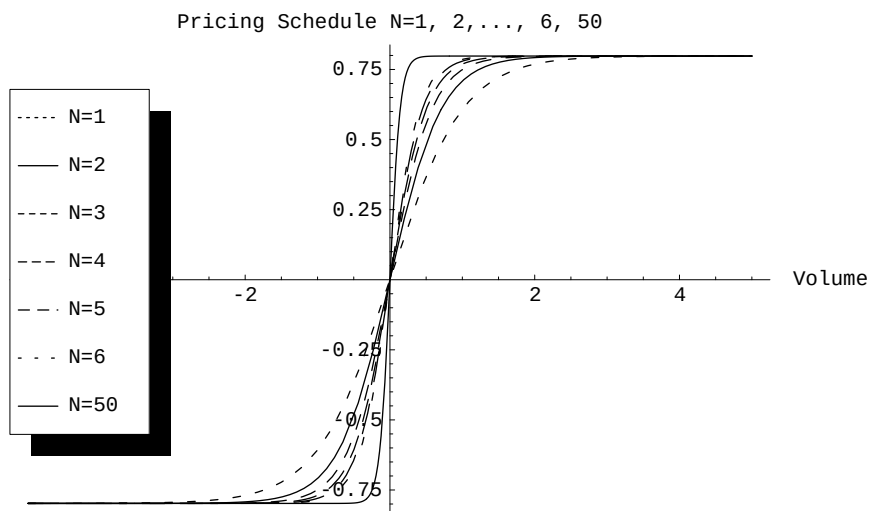


Figure 19: Pricing Schedule N .

to the semi strong efficient one p^* . For large enough wrongly directed noise trading, since the market makers belief is rather of the bearish state, we obtain the worst price $-p^*$.

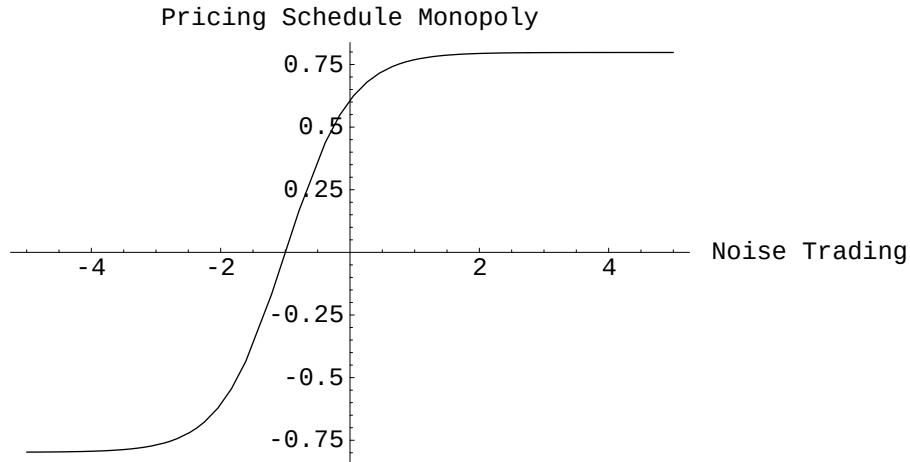


Figure 20: Pricing Schedule $\mathbf{N} = 1$ Bullish State.

Figure 21 shows the impact and consequences of competition on the welfare. We have plotted the pricing schedule in the bullish state of the risky asset as a function of the noise trading for the duopoly case (dash line) and the monopoly case (dotted line), given that the aggregate volumes submitted by the informed agents are respectively $2\gamma^*(2)$ and $\gamma^*(1)$.

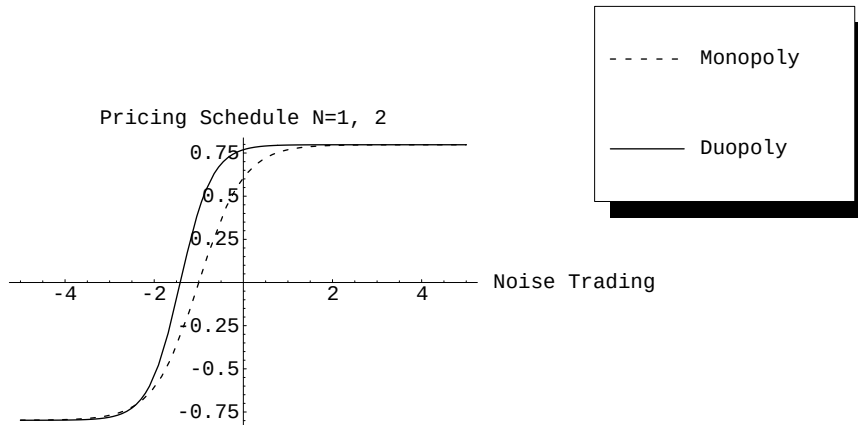


Figure 21: Pricing Schedule $\mathbf{N} = 1, 2$ Bullish State.

The efficiency gains from competition are two-fold:

- First, we see that the dash line is *almost always* above the dotted line. This exactly means that prices are more efficient in the duopoly case.
- Second, the range of noise trading, for which prices are in the correct direction, is larger in competition.

Of course, we observe, that for noise trading which is large enough in the opposite direction to the state of nature, that prices are uninformative. Moreover, we observe that in this range of noise trading, prices are even less efficient in the duopoly case (the dash line is below the dotted one). This has again to be related to the market makers beliefs of the bullish state of nature having observed a strongly negative aggregate volume in the two different situations. Indeed they affect a higher probability in the monopoly case than in the duopoly one because $2\gamma^*(2) > \gamma^*(1)$. However, the whole efficiency gains from competition should be roughly measured by the surface delineated by the two curves (dash and dotted ones)¹⁶. The sign is positive whenever the dash line is above and conversely negative whenever the dash line is below. We observe therefore an efficiency gain from competition.

These patterns are confirmed in figure 22 while examining several oligopolies.

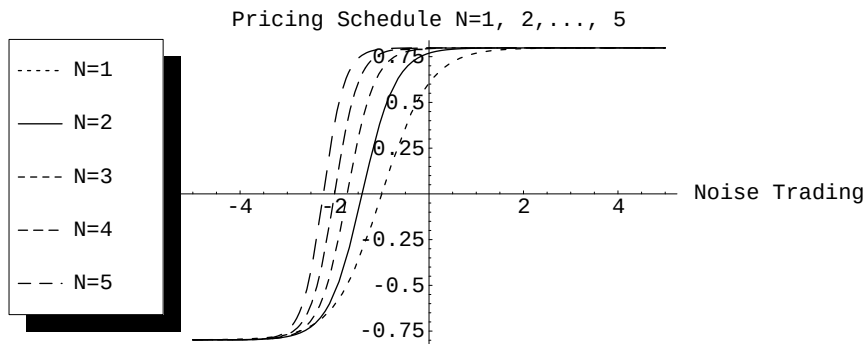


Figure 22: Pricing Schedule **N** Bullish State.

4 Concluding Remarks

We have proposed, in this paper, to study a financial market where risk neutral traders are endowed with bullish-bearish type of information. At the equilibrium, the optimal strategy

¹⁶Or more generally by the weighted surface according to the probability measure associated with the noise trading.

consists in buying (resp. selling) when bullish (resp. bearish) a fixed quantity. The pricing schedule is a non-linear function of the volumes. We have shown i) that the price function is a Sigmoid-shaped function. ii) The price pressure is a regular bell-shaped curve of the volumes and the liquidity is a convex U-shaped curve of the volumes. This is not in contradiction with the financial empirical studies. However to the best of our knowledge, no formal test of the link between the concave shape of the prices and the bounded information (namely bullish bearish signals) has been so far proposed. This could be done while using the dataset of Kim, Lin and Slovin (1997) who are able to precisely identify the time when the information is private from the time when it goes public. iii) A monopolistic bullish-bearish trader makes nearly thirty six percent of the profits she would have made with a perfect signal in a linear model à la Kyle (1985). iv) In the presence of competition, the market reveals the private information quicker than in a perfect informed strategic oligopoly.

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Appendices

A Proof of propositions 2.1 – 2.2

The informed agent maximizes his conditional expected profit conditionally on the observed signal S . If the informed agent observes the value $S = 0$ (which occurs with probability 0), it is straightforward to show that he optimally submits 0. Given the structure of his information and the symmetry of the problem, his optimal strategy is of the form:

$$\begin{aligned}\tilde{x} &= \gamma, \quad \text{if } S = 1, \\ \tilde{x} &= -\gamma, \quad \text{if } S = -1.\end{aligned}$$

Given their prior, the market makers conjecture that the informed agent submits the following order:

$$\begin{aligned}\tilde{x} &= \gamma^c, \quad \text{if } S = 1, \\ \tilde{x} &= -\gamma^c, \quad \text{if } S = -1,\end{aligned}$$

and consequently set prices according to (4.1):

$$\begin{aligned}\tilde{p} &= E(\tilde{v}/w) = \frac{\sigma_v}{\sqrt{2\pi}} [K_2(w) - K_1(w)], \\ \varphi(x) &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right), \\ K_1(w) &= \frac{2\varphi\left(\frac{w+\gamma^c}{\sigma_u}\right)}{\varphi\left(\frac{w-\gamma^c}{\sigma_u}\right) + \varphi\left(\frac{w+\gamma^c}{\sigma_u}\right)}, \\ K_2(w) &= 2 - K_1(w).\end{aligned}\tag{4.1}$$

The proof proceeds in two steps. We first compute the joint density of $(\tilde{v}, \tilde{w})$: $f_{\tilde{v}, \tilde{w}}$ and therefore the conditional p.d.f. of $\tilde{v}$ given $\tilde{w}$. We then deduce the conditional expectation $E(\tilde{v}/w)$.

$$\begin{aligned}F_{\tilde{v}, \tilde{w}}(v, w) &= \text{Prob}[\tilde{v} \leq v, \tilde{w} \leq w], \\ &= \text{Prob}[\tilde{v} \leq v, \tilde{w} \leq w/\tilde{v} < 0] \text{Prob}[\tilde{v} < 0] \\ &\quad + \text{Prob}[\tilde{v} \leq v, \tilde{w} \leq w/\tilde{v} > 0] \text{Prob}[\tilde{v} > 0], \\ &= \text{Prob}[\tilde{v} \leq v, -\gamma^c + \tilde{u} \leq w/\tilde{v} < 0] \text{Prob}[\tilde{v} < 0] \\ &\quad + \text{Prob}[\tilde{v} \leq v, \gamma^c + \tilde{u} \leq w/\tilde{v} > 0] \text{Prob}[\tilde{v} > 0], \\ &= \frac{1}{2} \{ \text{Prob}[\tilde{v} \leq v, -\gamma^c + \tilde{u} \leq w/\tilde{v} < 0] + \text{Prob}[\tilde{v} \leq v, \gamma^c + \tilde{u} \leq w/\tilde{v} > 0] \}, \\ &= \frac{1}{2} \{ \text{Prob}[\tilde{v} \leq v/\tilde{v} < 0] \text{Prob}[-\gamma^c + \tilde{u} \leq w] + \text{Prob}[\tilde{v} \leq v/\tilde{v} > 0] \text{Prob}[\gamma^c + \tilde{u} \leq w] \}, \\ &= \text{Prob}[\tilde{v} \leq \min(v, 0)] \text{Prob}[-\gamma^c + \tilde{u} \leq w] + \text{Prob}[0 < \tilde{v} \leq v] \text{Prob}[\gamma^c + \tilde{u} \leq w], \\ &= \Phi\left(\frac{\min(v, 0)}{\sigma_v}\right) \Phi\left(\frac{w+\gamma^c}{\sigma_u}\right) + \left[\Phi\left(\frac{v}{\sigma_v}\right) - \Phi(0)\right] \mathbb{I}_{v>0} \Phi\left(\frac{w-\gamma^c}{\sigma_u}\right),\end{aligned}$$

where: $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp\left(-\frac{1}{2}t^2\right) dt = \int_{-\infty}^x \varphi(t) dt$,

$$\begin{aligned} F_{\tilde{v}, \tilde{w}}(v, w) &= \Phi\left(\frac{v}{\sigma_v}\right) \Phi\left(\frac{w + \gamma^c}{\sigma_u}\right) \mathbb{I}_{v \leq 0} + \mathbb{I}_{v > 0} \frac{1}{2} \Phi\left(\frac{w + \gamma^c}{\sigma_u}\right) \\ &\quad + \Phi\left(\frac{v}{\sigma_v}\right) \Phi\left(\frac{w - \gamma^c}{\sigma_u}\right) \mathbb{I}_{v > 0} - \frac{1}{2} \mathbb{I}_{v > 0} \Phi\left(\frac{w - \gamma^c}{\sigma_u}\right), \end{aligned}$$

$$\begin{aligned} F_{\tilde{v}, \tilde{w}}(v, w) &= \Phi\left(\frac{v}{\sigma_v}\right) \left[\Phi\left(\frac{w + \gamma^c}{\sigma_u}\right) \mathbb{I}_{v \leq 0} + \Phi\left(\frac{w - \gamma^c}{\sigma_u}\right) \mathbb{I}_{v > 0} \right] \\ &\quad + \frac{1}{2} \mathbb{I}_{v > 0} \left[\Phi\left(\frac{w + \gamma^c}{\sigma_u}\right) - \Phi\left(\frac{w - \gamma^c}{\sigma_u}\right) \right]. \end{aligned}$$

If $v < 0$:

$$F_{\tilde{v}, \tilde{w}}(v, w) = \Phi\left(\frac{v}{\sigma_v}\right) \Phi\left(\frac{w + \gamma^c}{\sigma_u}\right),$$

$$f_{\tilde{v}, \tilde{w}}(v, w) = \frac{1}{\sigma_v \sigma_u} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w + \gamma^c}{\sigma_u}\right).$$

If $v > 0$:

$$F_{\tilde{v}, \tilde{w}}(v, w) = \Phi\left(\frac{v}{\sigma_v}\right) \Phi\left(\frac{w - \gamma^c}{\sigma_u}\right) + \frac{1}{2} \left[\Phi\left(\frac{w + \gamma^c}{\sigma_u}\right) - \Phi\left(\frac{w - \gamma^c}{\sigma_u}\right) \right],$$

$$f_{\tilde{v}, \tilde{w}}(v, w) = \frac{1}{\sigma_u \sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w - \gamma^c}{\sigma_u}\right).$$

Moreover $F_{\tilde{v}, \tilde{w}}(w) = \frac{1}{2} \left[\Phi\left(\frac{w + \gamma^c}{\sigma_u}\right) + \Phi\left(\frac{w - \gamma^c}{\sigma_u}\right) \right]$, therefore:

$$f_{\tilde{w}}(w) = \frac{1}{2\sigma_u} \left[\varphi\left(\frac{w + \gamma^c}{\sigma_u}\right) + \varphi\left(\frac{w - \gamma^c}{\sigma_u}\right) \right].$$

We also have by Bayes rule that:

$$f_{\tilde{v}/\tilde{w}}(v, w) = \frac{f_{\tilde{v}, \tilde{w}}(v, w)}{f_{\tilde{w}}(w)},$$

and thus:

$$\begin{aligned} \bullet \text{ if } v < 0, \quad f_{\tilde{v}/\tilde{w}}(v, w) &= \frac{\frac{2}{\sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w + \gamma^c}{\sigma_u}\right)}{\varphi\left(\frac{w + \gamma^c}{\sigma_u}\right) + \varphi\left(\frac{w - \gamma^c}{\sigma_u}\right)}, \\ \bullet \text{ if } v > 0, \quad f_{\tilde{v}/\tilde{w}}(v, w) &= \frac{\frac{2}{\sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w - \gamma^c}{\sigma_u}\right)}{\varphi\left(\frac{w + \gamma^c}{\sigma_u}\right) + \varphi\left(\frac{w - \gamma^c}{\sigma_u}\right)}. \end{aligned}$$

We define:

$$K_1(w) = \frac{2\varphi\left(\frac{w + \gamma^c}{\sigma_u}\right)}{\varphi\left(\frac{w + \gamma^c}{\sigma_u}\right) + \varphi\left(\frac{w - \gamma^c}{\sigma_u}\right)},$$

$$K_2(w) = \frac{2\varphi\left(\frac{w - \gamma^c}{\sigma_u}\right)}{\varphi\left(\frac{w + \gamma^c}{\sigma_u}\right) + \varphi\left(\frac{w - \gamma^c}{\sigma_u}\right)}.$$

Of course $K_1(w) + K_2(w) = 2$.

$$\begin{aligned} f_{\tilde{v}/\tilde{w}}(v, w) &= \frac{1}{\sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) [K_1(w) \mathbb{I}_{v < 0} + K_2(w) \mathbb{I}_{v > 0}] \cdot \\ E[\tilde{v}/w] &= \int_{-\infty}^{0^-} \frac{1}{\sigma_v} v \varphi\left(\frac{v}{\sigma_v}\right) K_1(w) dv + \int_{0^+}^{+\infty} \frac{1}{\sigma_v} v \varphi\left(\frac{v}{\sigma_v}\right) K_2(w) dv \cdot \\ E[\tilde{v}/w] &= \frac{\sigma_v}{\sqrt{2\pi}} [K_2(w) - K_1(w)]. \end{aligned}$$

The expected profit of the informed agent $\pi(\gamma, \gamma^c)$ is:

$$\pi(\gamma, \gamma^c) = E[\gamma(\tilde{v} - \tilde{p})/\tilde{v} > 0] \text{Prob}[\tilde{v} > 0] + E[-\gamma(\tilde{v} - \tilde{p})/\tilde{v} \leq 0] \text{Prob}[\tilde{v} \leq 0],$$

$$\pi(\gamma, \gamma^c) = E[\gamma(\tilde{v} - \tilde{p})/\tilde{v} > 0],$$

since $\text{Prob}[\tilde{v} \geq 0] = \text{Prob}[\tilde{v} < 0] = \frac{1}{2}$, we have by symmetry:

$$E[\gamma(\tilde{v} - \tilde{p})/\tilde{v} > 0] = E[-\gamma(\tilde{v} - \tilde{p})/\tilde{v} < 0],$$

$$\tilde{p} = \frac{2\sigma_v}{\sqrt{2\pi}} [K_2(w) - 1].$$

$$f_{\tilde{v}, \tilde{w}/\tilde{v} > 0}(v, w) = \frac{2}{\sigma_u \sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w - \gamma}{\sigma_u}\right) \mathbb{I}_{v > 0}.$$

$$\Rightarrow \pi(\gamma, \gamma^c) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \gamma(v - 2\frac{\sigma_v}{\sqrt{2\pi}} [K_2(w) - 1]) \frac{2}{\sigma_u \sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w - \gamma}{\sigma_u}\right) \mathbb{I}_{v > 0} dv dw,$$

$$\pi(\gamma, \gamma^c) = \frac{2\gamma\sigma_v}{\sqrt{2\pi}} \left\{ 1 - \int_{-\infty}^{+\infty} \frac{1}{\sigma_u} \varphi\left(\frac{w - \gamma}{\sigma_u}\right) [K_2(w) - 1] dw \right\},$$

$$\pi(\gamma, \gamma^c) = \frac{2\gamma\sigma_v}{\sqrt{2\pi}} \left\{ 2 - \int_{-\infty}^{+\infty} \frac{1}{\sigma_u} \varphi\left(\frac{w - \gamma}{\sigma_u}\right) K_2(w) dw \right\},$$

$$\pi(\gamma, \gamma^c) = \frac{2\gamma\sigma_v}{\sqrt{2\pi}} \left\{ 2 - \int_{-\infty}^{\infty} \frac{1}{\sigma_u} \varphi\left(\frac{w - \gamma}{\sigma_u}\right) K_2(w) dw \right\},$$

$$\pi(\gamma, \gamma^c) = \frac{2\gamma\sigma_v}{\sqrt{2\pi}} \left\{ 2 - \int_{-\infty}^{+\infty} \varphi(t) K_2(t\sigma_u + \gamma) dt \right\},$$

$$\pi(\gamma, \gamma^c) = \frac{2\gamma\sigma_v}{\sqrt{2\pi}} \left\{ 2 - \int_{-\infty}^{+\infty} \frac{2\varphi\left(t + \frac{\gamma - \gamma^c}{\sigma_u}\right) \varphi(t)}{\varphi\left(t + \frac{\gamma + \gamma^c}{\sigma_u}\right) + \varphi\left(t + \frac{\gamma - \gamma^c}{\sigma_u}\right)} dt \right\},$$

$$\pi(\gamma, \gamma^c) = \frac{2\gamma\sigma_v}{\sqrt{2\pi}} \left\{ 2 - \int_{-\infty}^{+\infty} \frac{2\varphi\left(t + \frac{\gamma - \gamma^c}{\sigma_u}\right) \varphi(t)}{\varphi\left(t + \frac{\gamma + \gamma^c}{\sigma_u}\right) + \varphi\left(t + \frac{\gamma - \gamma^c}{\sigma_u}\right)} dt \right\},$$

$$\pi(\gamma, \gamma^c) = \frac{4\gamma\sigma_v}{\sqrt{2\pi}} \left\{ 1 - \int_{-\infty}^{+\infty} \frac{\varphi\left(t + \frac{\gamma - \gamma^c}{\sigma_u}\right) \varphi(t)}{\varphi\left(t + \frac{\gamma + \gamma^c}{\sigma_u}\right) + \varphi\left(t + \frac{\gamma - \gamma^c}{\sigma_u}\right)} dt \right\},$$

$$\pi(\gamma, \gamma^c) = \frac{4\gamma\sigma_v}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \frac{\varphi\left(t + \frac{\gamma + \gamma^c}{\sigma_u}\right) \varphi(t)}{\varphi\left(t + \frac{\gamma + \gamma^c}{\sigma_u}\right) + \varphi\left(t + \frac{\gamma - \gamma^c}{\sigma_u}\right)} dt,$$

$\pi(0, \gamma^c) = \pi(+\infty, \gamma^c) = 0$, therefore there exists $\gamma^*(\gamma^c) > 0$ such that:

$$\gamma^*(\gamma^c) = \underset{\gamma \geq 0}{\operatorname{argmax}} \pi(\gamma, \gamma^c) = \underset{\gamma > 0}{\operatorname{argmax}} \gamma \int_{-\infty}^{+\infty} \frac{\varphi\left(t + \frac{\gamma + \gamma^c}{\sigma_u}\right) \varphi(t)}{\varphi\left(t + \frac{\gamma + \gamma^c}{\sigma_u}\right) + \varphi\left(t + \frac{\gamma - \gamma^c}{\sigma_u}\right)} dt,$$

We denote:

$$\gamma = \sigma_u \mu,$$

$$\gamma^c = \sigma_u \mu_c.$$

At the equilibrium, $\frac{\partial \pi}{\partial \mu}(\mu^*, \mu_c^*) = 0$ and $\mu^* = \mu_c^*$.

$$\begin{aligned} \pi(\mu, \mu_c) &= \frac{2\sigma_u\sigma_v}{\pi} \mu \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t + 2\mu_c)^2\right) \exp(2\mu_c(\mu - \mu_c))}, \\ \frac{\partial \pi}{\partial \mu}(\mu, \mu_c) &= \frac{2\sigma_u\sigma_v}{\pi} \left\{ \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t + 2\mu_c)^2\right) \exp(2\mu_c(\mu - \mu_c))} - \right. \\ &\quad \left. \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}(t + 2\mu_c)^2\right) \exp(2\mu_c(\mu - \mu_c)) 2\mu_c dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t + 2\mu_c)^2\right) \exp(2\mu_c(\mu - \mu_c))\right]^2} \right\}. \end{aligned}$$

At the equilibrium, we have:

$$\int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t + 2\mu)^2\right)} - 2\mu^2 \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}(t + 2\mu)^2\right) dt}{\left\{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t + 2\mu)^2\right)\right\}^2} = 0.$$

In other words we have:

$$H(\mu) - K(\mu) = 0,$$

$$H(\mu) = \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t + 2\mu)^2\right)},$$

$$K(\mu) = 2\mu^2 \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}(t + 2\mu)^2\right) dt}{\left\{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t + 2\mu)^2\right)\right\}^2}.$$

We now show that $K(\mu) = \mu^2 H(\mu)$:

$$\begin{aligned}
\frac{K(\mu)}{\mu^2} - H(\mu) &= \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}(t+2\mu)^2\right) - \exp\left(\frac{1}{2}t^2\right)}{\left[\exp\left(\frac{1}{2}(t+2\mu)^2\right) + \exp\left(\frac{1}{2}t^2\right)\right]^2} dt, \\
&= \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}u^2\right)}{\left[\exp\left(\frac{1}{2}(u-2\mu)^2\right) + \exp\left(\frac{1}{2}u^2\right)\right]^2} du - \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}t^2\right)}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2\mu)^2\right)\right]^2} dt, \\
&= \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}v^2\right)}{\left[\exp\left(\frac{1}{2}v^2\right) + \exp\left(\frac{1}{2}(v+2\mu)^2\right)\right]^2} dv - \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}t^2\right)}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2\mu)^2\right)\right]^2} dt = 0.
\end{aligned}$$

Thus:

$$H(\mu) - K(\mu) = 0,$$

$$\iff H(\mu)(1 - \mu^2) = 0.$$

Given that $H(\mu) > 0$, we obtain the solution $\mu = 1$. Concerning the deviations issues, we need to show that $\frac{\partial^2 \pi}{\partial \mu^2}(1, 1) < 0$.

$$\begin{aligned}
\frac{\partial^2 \pi}{\partial \mu^2}(\mu, \mu_c) &= \frac{2\sigma_u \sigma_v}{\pi} \left\{ - \int_{-\infty}^{+\infty} \frac{4\mu_c \exp\left(\frac{1}{2}(t+2\mu_c)^2\right) \exp(2\mu_c(\mu - \mu_c)) dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2\mu_c)^2\right) \exp(2\mu_c(\mu - \mu_c))\right]^2} + \right. \\
&\quad \int_{-\infty}^{+\infty} \frac{\left[\exp\left(\frac{1}{2}(t+2\mu_c)^2\right)\right]^2 8\mu_c^2 \exp(4\mu_c(\mu - \mu_c)) dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2\mu_c)^2\right) \exp(2\mu_c(\mu - \mu_c))\right]^3} - \\
&\quad \left. \int_{-\infty}^{+\infty} \frac{\left[\exp\left(\frac{1}{2}(t+2\mu_c)^2\right)\right]^2 \exp(2\mu_c(\mu - \mu_c)) 4\mu_c^2 \mu dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2\mu_c)^2\right) \exp(2\mu_c(\mu - \mu_c))\right]^2} \right\}.
\end{aligned}$$

$$\mu = \mu_c = 1.$$

$$\begin{aligned}
\frac{\partial^2 \pi}{\partial \mu^2}(1, 1) &= \frac{2\sigma_u \sigma_v}{\pi} \left\{ - \int_{-\infty}^{+\infty} \frac{4 \exp\left(\frac{1}{2}(t+2)^2\right) dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2)^2\right)\right]^2} + \right. \\
&\quad \left. \int_{-\infty}^{+\infty} \frac{8 \left[\exp\left(\frac{1}{2}(t+2)^2\right)\right]^2 dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2)^2\right)\right]^3} - \int_{-\infty}^{+\infty} \frac{4 \exp\left(\frac{1}{2}(t+2)^2\right) dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2)^2\right)\right]^2} \right\},
\end{aligned}$$

$$\frac{\partial^2 \pi}{\partial \mu^2}(1, 1) =$$

$$\frac{2\sigma_u\sigma_v}{\pi} \left\{ -8 \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}(t+2)^2\right) dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2)^2\right)\right]^2} + 8 \int_{-\infty}^{+\infty} \frac{\exp\left((t+2)^2\right) dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2)^2\right)\right]^3} \right\},$$

$$\frac{\partial^2 \pi}{\partial \mu^2}(1, 1) = -\frac{16\sigma_u\sigma_v}{\pi} \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}(t+2)^2\right) \exp\left(\frac{1}{2}t^2\right) dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2)^2\right)\right]^3} < 0.$$

$$\pi(1, 1) = \frac{2\sigma_u\sigma_v}{\pi} \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2)^2\right)} = \pi_{kyle} \xi^*,$$

with $\pi_{kyle} = \frac{1}{2}\sigma_u\sigma_v$ and $\xi^* = \frac{4}{\pi} \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t+2)^2\right)}$. This ends the proof of propositions 2.1 – 2.2.

B Proof of propositions 2.3 – 2.4

The informed agent i maximizes his conditional expected profit conditionally on the observed signal S . If the informed agent i observes the value $S = 0$ (which occurs with probability 0), it is straightforward to show that he optimally submits 0. Given the structure of his information and the symmetry of the problem, his optimal strategy is of the form:

$$\tilde{x}_i = \gamma_i, \quad \text{if } S = 1,$$

$$\tilde{x}_i = -\gamma_i, \quad \text{if } S = -1.$$

Given their prior, the market makers conjecture that the informed agent i submits the following order:

$$\tilde{x} = \gamma_i^c, \quad \text{if } S = 1,$$

$$\tilde{x}_i = -\gamma_i^c, \quad \text{if } S = -1,$$

The market makers set prices according to (4.2):

$$\begin{aligned}
\tilde{p} &= E[\tilde{v}/w] = \frac{\sigma_v}{\sqrt{2\pi}}[K_2(w, N) - K_1(w, N)], \\
\text{where :} \\
\varphi(x) &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right), \\
K_1(w, N) &= \frac{2\varphi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right)}{\varphi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) + \varphi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right)}, \\
K_2(w, N) &= 2 - K_1(w, N).
\end{aligned} \tag{4.2}$$

The proof proceeds in two steps. We first compute the joint density of $(\tilde{v}, \tilde{w}) : f_{\tilde{v}, \tilde{w}}$ and therefore the conditional p.d.f. of $\tilde{v}$ given w . We thus deduce the conditional expectation $E[\tilde{v}/w]$.

$$\begin{aligned}
F_{\tilde{v}, \tilde{w}}(v, w) &= \text{Prob}[\tilde{v} \leq v, \tilde{w} \leq w], \\
&= \text{Prob}[\tilde{v} \leq v, \tilde{w} \leq w/\tilde{v} < 0] \text{Prob}[\tilde{v} < 0] + \text{Prob}[\tilde{v} \leq v, \tilde{w} \leq w/\tilde{v} > 0] \text{Prob}[\tilde{v} > 0], \\
&= \frac{1}{2} \left\{ \text{Prob}[\tilde{v} \leq v, -\sum_{i=1}^N \gamma_i^c + \tilde{u} \leq w/\tilde{v} < 0] + \text{Prob}[\tilde{v} \leq v, \sum_{i=1}^N \gamma_i^c + \tilde{u} \leq w/\tilde{v} > 0] \right\}, \\
&= \frac{1}{2} \left\{ \text{Prob}\left[\tilde{v} \leq v, -\sum_{i=1}^N \gamma_i^c + \tilde{u} \leq w/\tilde{v} < 0\right] + \text{Prob}\left[\tilde{v} \leq v, \sum_{i=1}^N \gamma_i^c + \tilde{u} \leq w/\tilde{v} > 0\right] \right\}, \\
&= \text{Prob}\left[\tilde{v} \leq \min(0, v), -\sum_{i=1}^N \gamma_i^c + \tilde{u} \leq w\right] + \text{Prob}\left[0 < \tilde{v} \leq v, \sum_{i=1}^N \gamma_i^c + \tilde{u} \leq w\right], \\
&= \Phi\left(\frac{\min(0, v)}{\sigma_v}\right) \Phi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) + \left[\Phi\left(\frac{v}{\sigma_v}\right) - \Phi(0)\right] \mathbb{I}_{v>0} \Phi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right),
\end{aligned}$$

$$F_{\tilde{v}, \tilde{w}}(v, w) =$$

$$\begin{aligned}
&\Phi\left(\frac{v}{\sigma_v}\right) \Phi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) \mathbb{I}_{v \leq 0} + \mathbb{I}_{v > 0} \frac{1}{2} \Phi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) + \Phi\left(\frac{v}{\sigma_v}\right) \Phi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) \mathbb{I}_{v > 0} - \frac{1}{2} \mathbb{I}_{v > 0} \Phi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right), \\
&= \Phi\left(\frac{v}{\sigma_v}\right) \left[\Phi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) \mathbb{I}_{v \leq 0} + \Phi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) \mathbb{I}_{v > 0} \right] + \frac{1}{2} \mathbb{I}_{v > 0} \left[\Phi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) - \Phi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) \right].
\end{aligned}$$

If $v < 0$:

$$F_{\tilde{v}, \tilde{w}}(v, w) = \Phi\left(\frac{v}{\sigma_v}\right) \Phi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right),$$

$$f_{\tilde{v}, \tilde{w}}(v, w) = \frac{1}{\sigma_u \sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right).$$

If $v > 0$:

$$F_{\tilde{v}, \tilde{w}}(v, w) = \Phi\left(\frac{v}{\sigma_v}\right) \Phi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) + \frac{1}{2} \left[\Phi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) - \Phi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) \right],$$

$$f_{\tilde{v}, \tilde{w}}(v, w) = \frac{1}{\sigma_u \sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right).$$

Moreover $F_{\tilde{w}}(w) = \frac{1}{2} \left[\Phi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) + \Phi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) \right]$, therefore:

$$f_{\tilde{w}}(w) = \frac{1}{2\sigma_u} \left[\varphi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) + \varphi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) \right]. \text{ We also have, applying Bayes rule, that:}$$

$$f_{\tilde{v}/\tilde{w}}(v, w) = \frac{f_{\tilde{v}, \tilde{w}}(v, w)}{f_{\tilde{w}}(w)},$$

and thus:

$$\bullet \text{ if } v < 0, \quad f_{\tilde{v}/\tilde{w}}(v, w) = \frac{2}{\sigma_v} \frac{\varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right)}{\varphi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) + \varphi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right)},$$

$$\bullet \text{ if } v > 0, \quad f_{\tilde{v}/\tilde{w}}(v, w) = \frac{2}{\sigma_v} \frac{\varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right)}{\varphi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) + \varphi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right)}.$$

We define:

$$K_1(w, N) = \frac{2\varphi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right)}{\varphi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) + \varphi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right)},$$

$$K_2(w, N) = \frac{2\varphi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right)}{\varphi\left(\frac{w - \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right) + \varphi\left(\frac{w + \sum_{i=1}^N \gamma_i^c}{\sigma_u}\right)}.$$

Of course $K_1(w, N) + K_2(w, N) = 2$.

$$f_{\tilde{v}/\tilde{w}}(v, w) = \frac{1}{\sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) [K_1(w, N) \mathbb{I}_{v < 0} + K_2(w, N) \mathbb{I}_{v > 0}],$$

$$E[\tilde{v}/w] = \int_{-\infty}^{0^-} \frac{v}{\sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) K_1(w, N) dv + \int_{0^+}^{+\infty} \frac{v}{\sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) K_2(w, N) dv,$$

$$E[\tilde{v}/w] = \frac{\sigma_v}{\sqrt{2\pi}} [K_2(w, N) - K_1(w, N)].$$

We define $\gamma = (\gamma_1, \dots, \gamma_N)'$ and $\gamma_c = (\gamma_1^c, \dots, \gamma_N^c)'$. The expected profit $\pi_i(\gamma, \gamma_c)$ of agent i is:

$$\pi_i(\gamma, \gamma_c) = E[\gamma_i(\tilde{v} - \tilde{p})/\tilde{v} > 0] Prob[\tilde{v} > 0] + E[-\gamma_i(\tilde{v} - \tilde{p})/\tilde{v} < 0] Prob[\tilde{v} \leq 0],$$

$$= \frac{1}{2} \{E[\gamma_i(\tilde{v} - \tilde{p})/\tilde{v} > 0] + E[-\gamma_i(\tilde{v} - \tilde{p})/\tilde{v} < 0]\},$$

$$= \frac{1}{2} \left\{ \int_0^{+\infty} \int_{-\infty}^{+\infty} \gamma_i(v - \tilde{p}) \frac{2}{\sigma_u \sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w - \sum_{i=1}^N \gamma_i}{\sigma_u}\right) dv dw \right. \\ \left. + \int_{-\infty}^0 \int_{-\infty}^{+\infty} -\gamma_i(v - \tilde{p}) \frac{2}{\sigma_u \sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w + \sum_{i=1}^N \gamma_i}{\sigma_u}\right) dv dw \right\}.$$

We make the following change in variables for the second integral $\begin{pmatrix} v \\ w \end{pmatrix} \rightarrow \begin{pmatrix} -v \\ -w \end{pmatrix}$ while taking into account that $\varphi(x) = \varphi(-x)$, $P(-w) = -P(w)$:

$$\begin{aligned}
\pi_i(\gamma, \gamma_c) &= \frac{1}{2} \left\{ \int_0^{+\infty} \int_{-\infty}^{+\infty} \gamma_i(v - \tilde{p}) \frac{2}{\sigma_u \sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w - \sum_{i=1}^N \gamma_i}{\sigma_u}\right) dv dw \right. \\
&\quad \left. + \int_0^{+\infty} \int_{-\infty}^{+\infty} \gamma_i(v - \tilde{p}) \frac{2}{\sigma_u \sigma_v} \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w - \sum_{i=1}^N \gamma_i}{\sigma_u}\right) dv dw \right\}, \\
&= \frac{2\gamma_i}{\sigma_u \sigma_v} \int_0^{+\infty} \int_{-\infty}^{+\infty} (v - \tilde{p}) \varphi\left(\frac{v}{\sigma_v}\right) \varphi\left(\frac{w - \sum_{i=1}^N \gamma_i}{\sigma_u}\right) dv dw, \\
&= \frac{2\gamma_i}{\sigma_v} \int_0^{+\infty} \left[v - \int_{-\infty}^{+\infty} 2 \frac{\sigma_v}{\sqrt{2\pi}} (K_2(w, N) - 1) \frac{\varphi\left(\frac{w - \sum_{i=1}^N \gamma_i}{\sigma_u}\right)}{\sigma_u} dw \right] \varphi\left(\frac{v}{\sigma_v}\right) dv, \\
&= \frac{2\gamma_i}{\sigma_v} \left[\frac{\sigma_v^2}{\sqrt{2\pi}} - \frac{\sigma_v^2}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} (K_2(w, N) - 1) \frac{\varphi\left(\frac{w - \sum_{i=1}^N \gamma_i}{\sigma_u}\right)}{\sigma_u} dw \right], \\
&= \frac{2\gamma_i \sigma_v}{\sqrt{2\pi}} \left[1 - \int_{-\infty}^{+\infty} [K_2(t\sigma_u + \sum_{i=1}^N \gamma_i, N) - 1] \varphi(t) dt \right], \\
&= \frac{4\gamma_i \sigma_v}{\sqrt{2\pi}} \left[1 - \frac{1}{2} \int_{-\infty}^{+\infty} K_2(t\sigma_u + \sum_{i=1}^N \gamma_i, N) \varphi(t) dt \right], \\
&= \frac{4\gamma_i \sigma_v}{\sqrt{2\pi}} \left[1 - \int_{-\infty}^{+\infty} \frac{\varphi\left(t + \frac{\sum_{i=1}^N \gamma_i - \gamma_i^c}{\sigma_u}\right)}{\varphi\left(t + \frac{\sum_{i=1}^N \gamma_i - \gamma_i^c}{\sigma_u}\right) + \varphi\left(t + \frac{\sum_{i=1}^N \gamma_i + \gamma_i^c}{\sigma_u}\right)} \varphi(t) dt \right].
\end{aligned}$$

We define $\gamma_i = \mu_i \sigma_u$, $\gamma_i^c = \mu_i^c \sigma_u$ and according $\mu = (\mu_1, \dots, \mu_N)'$, $\mu^c = (\mu_1^c, \dots, \mu_N^c)'$:

$$\pi_i(\mu, \mu_c) = \frac{4\mu_i \sigma_u \sigma_v}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \frac{\varphi\left(t + \sum_{i=1}^N (\mu_i + \mu_i^c)\right)}{\varphi\left(t + \sum_{i=1}^N (\mu_i - \mu_i^c)\right) + \varphi\left(t + \sum_{i=1}^N (\mu_i + \mu_i^c)\right)} \varphi(t) dt,$$

$$\pi_i(\mu, \mu_c) = \frac{2\sigma_u \sigma_v}{\pi} \mu_i \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}\left(t + 2\sum_{i=1}^N \mu_i^c\right)^2\right) \exp\left(2\sum_{i=1}^N \mu_i^c \sum_{i=1}^N (\mu_i - \mu_i^c)\right)},$$

At the value $\mu_i = 0$ and when μ_i tends towards infinity the value of the expected profit is zero, therefore there exists an interior solution $\mu_i^*(\mu_{-i}, \mu_c)$.

$$\frac{\partial \pi_i}{\partial \mu_i}(\mu, \mu_c) = \frac{2\sigma_u \sigma_v}{\pi} \left\{ \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}\left(t + 2\sum_{i=1}^N \mu_i^c\right)^2\right) \exp\left(2\sum_{i=1}^N \mu_i^c \sum_{i=1}^N (\mu_i - \mu_i^c)\right)} - \right.$$

$$\left. \mu_i \int_{-\infty}^{+\infty} \frac{\exp\left(\frac{1}{2}\left(t + 2\sum_{i=1}^N \mu_i^c\right)^2\right) 2\sum_{i=1}^N \mu_i^c \exp\left(2\sum_{i=1}^N \mu_i^c \sum_{i=1}^N (\mu_i - \mu_i^c)\right) dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}\left(t + 2\sum_{i=1}^N \mu_i^c\right)^2\right) \exp\left(2\sum_{i=1}^N \mu_i^c \sum_{i=1}^N (\mu_i - \mu_i^c)\right)\right]^2} \right\}.$$

Moreover at the equilibrium, we have:

$$\frac{\partial \pi_i}{\partial \mu_i}(\mu, \mu_c) = 0,$$

$$\mu_i^c = \mu_i, \quad i = 1, \dots, N,$$

This implies that:

$$\int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}\left(t + 2\sum_{i=1}^N \mu_i\right)^2\right)} - \mu_i \int_{-\infty}^{+\infty} \frac{2\sum_{i=1}^N \mu_i \exp\left(\frac{1}{2}\left(t + 2\sum_{i=1}^N \mu_i\right)^2\right) dt}{\left[\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}\left(t + 2\sum_{i=1}^N \mu_i\right)^2\right)\right]^2} = 0,$$

Thus for all $i = 1, \dots, N$, we have $\mu_i = \mu^*(N)$ independent of i , with:

$$H(N\mu^*(N)) - \frac{1}{N}K(N\mu^*(N)) = 0,$$

$$\iff H(N\mu^*(N)) \left(1 - N\mu^{*2}(N)\right) = 0,$$

$$\iff \mu^*(N) = \frac{1}{\sqrt{N}},$$

where $H(\cdot)$ and $K(\cdot)$ have been previously defined. Concerning the deviations issues, we need to prove that $\frac{\partial^2 \pi}{\partial \mu_i^2} \left(\frac{1}{\sqrt{N}}, \dots, \frac{1}{\sqrt{N}}, \frac{1}{\sqrt{N}}, \dots, \frac{1}{\sqrt{N}} \right) < 0$.

$$\begin{aligned} \frac{\partial^2 \pi}{\partial \mu_i^2}(\mu, \mu_c) = & \frac{2\sigma_u \sigma_v}{\pi} \left\{ -4 \sum_{i=1}^N \mu_i^c \int_{-\infty}^{+\infty} \frac{\exp \left(\frac{1}{2} \left(t + 2 \sum_{i=1}^N \mu_i^c \right)^2 \right) \exp \left(2 \sum_{i=1}^N \mu_i^c \sum_{i=1}^N (\mu_i - \mu_i^c) \right) dt}{\left[\exp \left(\frac{1}{2} t^2 \right) + \exp \left(\frac{1}{2} \left(t + 2 \sum_{i=1}^N \mu_i^c \right)^2 \right) \exp \left(2 \sum_{i=1}^N \mu_i^c \sum_{i=1}^N (\mu_i - \mu_i^c) \right) \right]^2} - \right. \\ & 4 \mu_i \left(\sum_{i=1}^N \mu_i^c \right)^2 \int_{-\infty}^{+\infty} \frac{\exp \left(\frac{1}{2} \left(t + 2 \sum_{i=1}^N \mu_i^c \right)^2 \right) \exp \left(2 \sum_{i=1}^N \mu_i^c \sum_{i=1}^N (\mu_i - \mu_i^c) \right) dt}{\left[\exp \left(\frac{1}{2} t^2 \right) + \exp \left(\frac{1}{2} \left(t + 2 \sum_{i=1}^N \mu_i^c \right)^2 \right) \exp \left(2 \sum_{i=1}^N \mu_i^c \sum_{i=1}^N (\mu_i - \mu_i^c) \right) \right]^2} + \\ & \left. 8 \mu_i \left(\sum_{i=1}^N \mu_i^c \right)^2 \int_{-\infty}^{+\infty} \frac{\exp \left(\left(t + 2 \sum_{i=1}^N \mu_i^c \right)^2 \right) \exp \left(2 \sum_{i=1}^N \mu_i^c \sum_{i=1}^N (\mu_i - \mu_i^c) \right) dt}{\left[\exp \left(\frac{1}{2} t^2 \right) + \exp \left(\frac{1}{2} \left(t + 2 \sum_{i=1}^N \mu_i^c \right)^2 \right) \exp \left(2 \sum_{i=1}^N \mu_i^c \sum_{i=1}^N (\mu_i - \mu_i^c) \right) \right]^3} \right\}, \\ \frac{\partial^2 \pi}{\partial^2 \mu_i}(\mu^*(N), \mu^*(N)) = & \frac{2\sigma_u \sigma_v}{\pi} \left\{ -8\sqrt{N} \int_{-\infty}^{+\infty} \frac{\exp \left(\frac{1}{2} (t + 2\sqrt{N})^2 \right) dt}{\left[\exp \left(\frac{1}{2} t^2 \right) + \exp \left(\frac{1}{2} (t + 2\sqrt{N})^2 \right) \right]^2} + \right. \\ & \left. 8\sqrt{N} \int_{-\infty}^{+\infty} \frac{\exp \left((t + 2\sqrt{N})^2 \right) dt}{\left[\exp \left(\frac{1}{2} t^2 \right) + \exp \left(\frac{1}{2} (t + 2\sqrt{N})^2 \right) \right]^3} \right\}, \\ \frac{\partial^2 \pi}{\partial^2 \mu_i}(\mu^*(N), \mu^*(N)) = & -\frac{16\sigma_u \sigma_v \sqrt{N}}{\pi} \int_{-\infty}^{+\infty} \frac{\exp \left(\frac{1}{2} (t + 2\sqrt{N})^2 \right) \exp \left(\frac{1}{2} t^2 \right) dt}{\left[\exp \left(\frac{1}{2} t^2 \right) + \exp \left(\frac{1}{2} (t + 2\sqrt{N})^2 \right) \right]^3} < 0. \end{aligned}$$

This ends both proofs of existence and uniqueness.

$$\pi_i(\mu^*(N), \mu^*(N)) = \frac{2\sigma_u \sigma_v}{\pi \sqrt{N}} \int_{-\infty}^{+\infty} \frac{dt}{\exp \left(\frac{1}{2} t^2 \right) + \exp \left(\frac{1}{2} (t + 2\sqrt{N})^2 \right)}.$$

Proof of propositions 3.1 – 3.2, – 3.3 :

Profits:

The individual expected profit is given by:

$$\begin{aligned}
\pi^*(\mu^*(N), \mu^*(N)) &= \frac{2\sigma_u\sigma_v}{\pi\sqrt{N}} \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}t^2\right) + \exp\left(\frac{1}{2}(t + 2\sqrt{N})^2\right)}, \\
&= \frac{2\sigma_u\sigma_v}{\pi\sqrt{N}} \int_{-\infty}^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}(t - \sqrt{N})^2\right) + \exp\left(\frac{1}{2}(t + \sqrt{N})^2\right)}, \\
&= \frac{4\sigma_u\sigma_v}{\pi\sqrt{N}} \int_0^{+\infty} \frac{dt}{\exp\left(\frac{1}{2}(t - \sqrt{N})^2\right) + \exp\left(\frac{1}{2}(t + \sqrt{N})^2\right)}, \\
&= \frac{4\sigma_u\sigma_v}{\pi\sqrt{N}} \int_0^{+\infty} \frac{\exp\left(-\frac{1}{2}(t + \sqrt{N})^2\right) dt}{1 + \exp(-2t\sqrt{N})}, \\
&= \frac{4\sigma_u\sigma_v}{\pi\sqrt{N}} \theta(\sqrt{N}),
\end{aligned}$$

with $\theta(x) = \int_0^{+\infty} \frac{\exp\left(-\frac{1}{2}(t+x)^2\right)}{1 + \exp(-2tx)} dt$. For $x > 0$, we perform the following changes in variables: $u = tx$.

$$\theta(x) = \frac{1}{x} \int_0^{+\infty} \frac{\exp\left[-\frac{1}{2}\left(\frac{u}{x} + x\right)^2\right]}{1 + \exp(-2u)} du,$$

$$\theta(x) = \frac{1}{x} \int_0^{+\infty} \frac{\exp\left(-\frac{1}{2}x^2\right) \exp(-u) \exp\left(-\frac{1}{2}\frac{u^2}{x^2}\right)}{1 + \exp(-2u)} du,$$

$$\theta(x) = \frac{1}{x} \exp\left(-\frac{1}{2}x^2\right) \int_0^{+\infty} \frac{\exp(-u) \exp\left(-\frac{1}{2}\frac{u^2}{x^2}\right)}{1 + \exp(-2u)} du.$$

$$\theta(\sqrt{N}) \underset{N \rightarrow +\infty}{\sim} \frac{1}{\sqrt{N}} \exp\left(-\frac{1}{2}N\right) \int_0^{+\infty} \frac{\exp(-u)}{1 + \exp(-2u)} du = \frac{1}{\sqrt{N}} \exp\left(-\frac{1}{2}N\right) \int_0^1 \frac{dv}{1 + v^2},$$

$$\theta(\sqrt{N}) \underset{N \rightarrow +\infty}{\sim} \frac{1}{\sqrt{N}} \exp\left(-\frac{1}{2}N\right) \frac{\pi}{4},$$

therefore:

$$\pi_i^*(\mu^*(N), \mu^*(N)) \underset{N \rightarrow +\infty}{\sim} \frac{\sigma_u\sigma_v}{N} \exp\left(-\frac{1}{2}N\right),$$

$$N\pi_i^*(\mu^*(N), \mu^*(N)) \underset{N \rightarrow +\infty}{\sim} \sigma_u\sigma_v \exp\left(-\frac{1}{2}N\right).$$

Pricing Schedule:

$$P(w) = \frac{2\sigma_v}{\sqrt{2\pi}} [K_2^*(w, N) - 1],$$

$$K_2^*(w, N) = \frac{2\varphi\left(\frac{w}{\sigma_u} - \sqrt{N}\right)}{\varphi\left(\frac{w}{\sigma_u} + \sqrt{N}\right) + \varphi\left(\frac{w}{\sigma_u} - \sqrt{N}\right)},$$

$$P(w) = \frac{2\sigma_v}{\sqrt{2\pi}} \frac{\varphi\left(\frac{w}{\sigma_u} - \sqrt{N}\right) - \varphi\left(\frac{w}{\sigma_u} + \sqrt{N}\right)}{\varphi\left(\frac{w}{\sigma_u} + \sqrt{N}\right) + \varphi\left(\frac{w}{\sigma_u} - \sqrt{N}\right)},$$

$$P(w) = \frac{2\sigma_v}{\sqrt{2\pi}} \frac{\exp\left(-\frac{1}{2}\left(\frac{w}{\sigma_u} - \sqrt{N}\right)^2\right) - \exp\left(-\frac{1}{2}\left(\frac{w}{\sigma_u} + \sqrt{N}\right)^2\right)}{\exp\left(-\frac{1}{2}\left(\frac{w}{\sigma_u} - \sqrt{N}\right)^2\right) + \exp\left(-\frac{1}{2}\left(\frac{w}{\sigma_u} + \sqrt{N}\right)^2\right)},$$

thus we obtain the result:

$$P(w) \underset{N \rightarrow +\infty}{\sim} \frac{2\sigma_v}{\sqrt{2\pi}}, \quad \text{if } S = 1,$$

$$P(w) \underset{N \rightarrow +\infty}{\sim} -\frac{2\sigma_v}{\sqrt{2\pi}}, \quad \text{if } S = -1.$$

Liquidity:

$\lambda(w, N) = \frac{\partial p}{\partial w}(w)$ is the price pressure and corresponds to the inverse of the liquidity:

$$\lambda(w, N) = \frac{8\sigma_v}{\sqrt{2\pi}\sigma_u} \frac{\sqrt{N} \exp\left(\frac{2w\sqrt{N}}{\sigma_u}\right)}{\left[1 + \exp\left(\frac{2w\sqrt{N}}{\sigma_u}\right)\right]^2},$$

$$\lambda(w, N) \underset{N \rightarrow +\infty}{\sim} \frac{8\sigma_v}{\sqrt{2\pi}\sigma_u} \sqrt{N} \exp\left(\frac{2w\sqrt{N}}{\sigma_u}\right), \quad \text{if } S = -1,$$

$$\lambda(w, N) \underset{N \rightarrow +\infty}{\sim} \frac{8\sigma_v}{\sqrt{2\pi}\sigma_u} \sqrt{N} \exp\left(-\frac{2w\sqrt{N}}{\sigma_u}\right), \quad \text{if } S = 1,$$

$$\lambda(w, N) \underset{N \rightarrow +\infty}{\sim} \frac{8\sigma_v}{\sqrt{2\pi}\sigma_u} \sqrt{N} \exp(-2N) \exp\left(\frac{2u\sqrt{N}}{\sigma_u}\right), \quad \text{if } S = -1,$$

$$\lambda(w, N) \underset{N \rightarrow +\infty}{\sim} \frac{8\sigma_v}{\sqrt{2\pi}\sigma_u} \sqrt{N} \exp(-2N) \exp\left(-\frac{2u\sqrt{N}}{\sigma_u}\right), \quad \text{if } S = 1.$$